Report On

Business Insight Generation Using EDA

A PROJECT REPORT

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in partial fulfilment for the award of the degree of

BACHELOR OF ENGINEERING

IN

Computer Science & Engineering (CSE)

Submitted to:

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April 2026

BONAFIDE CERTIFICATE

Certified that this project report "Business Insight Generation Using EDA" is the bonafide work of Deepanshi (24BCS11431), Rituraj (24BCS10588), Shourya (24BCS10270), and Jiya (24BCS11043), who carried out the project work under my supervision during the academic year 2025-2026.

This project report has been prepared in partial fulfillment of the requirements for the award of the degree of Bachelor of Engineering in Computer Science & Engineering at Chandigarh University. The work is original and has not been submitted elsewhere for the award of any degree or diploma.

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DECLARATION

We hereby declare that the project report entitled "Business Insight Generation Using EDA" submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Engineering in Computer Science & Engineering at Chandigarh University is an authentic record of our own work carried out during the academic year 2025–2026 under the supervision and guidance of Mr. Vishal Dutt, Assistant Professor, Department of Computer Science & Engineering.

The matter embodied in this project has not been submitted to any other university or institution for the award of any degree or diploma. All the information, data, and results presented in this report have been collected, processed, and analyzed by us using standard academic and professional methodologies. Wherever data, ideas, or text from other sources have been used, they have been duly acknowledged in the report with proper citations and references.

We understand that any false statement in this declaration may lead to cancellation of the degree. We are fully responsible for the authenticity, integrity, and originality of the content presented in this report.

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ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our respected faculty guide, Mr. Vishal Dutt, Assistant Professor, Department of Computer Science & Engineering, Chandigarh University, for his invaluable guidance, continuous support, and constant encouragement throughout the development of this project on "Business Insight Generation Using EDA." His clear explanations, expert suggestions, and constructive feedback helped us understand the concepts of data analytics and Exploratory Data Analysis (EDA) in a deeper and more practical manner. His mentorship played a crucial role in shaping our approach towards analyzing data and generating meaningful business insights.

We are also highly thankful to the Department of Computer Science & Engineering, Chandigarh University, for providing us with the necessary academic resources, technical infrastructure, and a positive learning environment. The availability of well-equipped computer laboratories, licensed software tools, high-speed internet connectivity, and extensive academic library resources greatly contributed to the successful completion of this project. The department has always encouraged practical learning and project-based education, which helped us apply theoretical knowledge to real-world data analysis scenarios.

We would like to extend our sincere appreciation to all the teaching and non-teaching staff members of the Department of Computer Science & Engineering who directly or indirectly supported us during different stages of the project. Their guidance, cooperation, timely assistance, and words of encouragement helped us overcome various technical and conceptual challenges that arose during the course of this project.

We are deeply grateful to our fellow team members for their outstanding dedication, active cooperation, and excellent teamwork. Each member contributed their unique skills, ideas, and best efforts to achieve the common objectives of this project. The spirit of collaboration, mutual understanding, and shared responsibility helped us complete the project efficiently and effectively within the stipulated time frame.

This project has also been a transformative learning experience for all of us. It significantly enhanced our technical knowledge in the field of data analytics, improved our analytical thinking and problem-solving capabilities, and strengthened our understanding of real-world data challenges. Working with actual retail datasets and applying EDA techniques gave us the confidence and skills needed to handle data-driven tasks in professional settings.

Finally, we would like to express our heartfelt gratitude to our parents, family members, and well-wishers for their constant motivation, moral support, and encouragement throughout our academic journey. Their unwavering belief in our abilities has always been our greatest source of strength, inspiration, and determination.

ABSTRACT

In today's highly competitive and data-driven business environment, organizations across all industries generate enormous volumes of data from their daily operations including sales transactions, customer interactions, supply chain activities, and digital engagements. However, the mere availability of raw data does not automatically translate into actionable intelligence or business value. The true potential of data can only be realized when it is systematically analyzed and interpreted to extract meaningful patterns, trends, and insights that support strategic decision-making.

This project focuses on generating comprehensive and actionable business insights using Exploratory Data Analysis (EDA) applied to retail sales data. The primary objective of this project is to analyze a structured retail dataset containing information about products, customers, sales transactions, regional performance, and profitability metrics. Through systematic application of EDA techniques, this project aims to understand sales trends over time, identify profit patterns across product categories and geographic regions, evaluate product performance across different dimensions, and quantify the impact of discount strategies on overall business profitability.

The methodology followed in this project involves multiple well-defined stages. In the first stage, the raw dataset is imported and subjected to comprehensive data cleaning and preprocessing operations. This includes identifying and handling missing values through appropriate imputation techniques, detecting and removing duplicate records, standardizing data formats and data types, and resolving inconsistencies in categorical variables. These preprocessing steps ensure that the dataset is accurate, complete, and ready for meaningful analysis.

In the second stage, descriptive statistical analysis is performed to summarize the key characteristics of the dataset. Measures of central tendency such as mean and median, measures of dispersion such as standard deviation and variance, and measures of distribution shape such as skewness and kurtosis are computed for all numerical variables. This statistical summarization provides a foundational understanding of the dataset's overall structure and the typical ranges and distributions of key business metrics.

In the third stage, a wide variety of data visualization techniques are employed to represent the patterns, relationships, and distributions present in the data in a clear and visually intuitive manner. These visualizations include bar charts for comparing categorical performance, line graphs for tracking temporal trends, histograms for displaying frequency distributions, scatter plots for exploring bivariate relationships, box plots for identifying outliers and data spread, pie charts for showing proportional composition, and correlation heatmaps for revealing inter-variable relationships.

The analytical findings reveal several significant business insights. Certain product categories consistently generate higher sales volumes but exhibit variable profit margins depending on the discount levels applied. Regional analysis identifies geographic areas of strong and weak performance, enabling targeted business strategies. Customer segmentation analysis uncovers distinct behavioral patterns across different customer groups. The relationship between discount depth and profit margin reveals a clear negative correlation, highlighting the risks of aggressive discounting strategies. These insights collectively provide a solid empirical foundation for data-driven business decisions related to pricing, inventory management, regional marketing, and customer targeting. The project successfully demonstrates EDA as a powerful and accessible tool for transforming raw business data into strategic intelligence.

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CHAPTER 1: INTRODUCTION

1.1 Overview of Data Analytics

In today's digital era, organizations generate massive amounts of data through various activities such as sales transactions, customer interactions, online services, and business operations. This data contains valuable information that can help businesses understand their performance, identify market opportunities, and improve strategic decision-making. However, raw data in its original form is often unstructured, incomplete, and difficult to interpret without the application of systematic analytical techniques. Data analytics provides the methodological framework to transform raw data into meaningful and actionable information.

Data analytics refers to the comprehensive process of collecting, organizing, cleaning, analyzing, and interpreting data to discover patterns, trends, correlations, and insights. It involves the use of statistical techniques, mathematical algorithms, computational tools, and visualization methods to extract useful knowledge from structured and unstructured data sources. Businesses across all sectors use data analytics to improve operational efficiency, reduce costs, understand customer behavior, optimize marketing strategies, and make evidence-based decisions.

The field of data analytics encompasses four distinct but complementary types: descriptive analytics, diagnostic analytics, predictive analytics, and prescriptive analytics. Descriptive analytics focuses on summarizing historical data to understand what has happened. Diagnostic analytics investigates the causes behind observed outcomes, answering the question of why something happened. Predictive analytics uses statistical models and machine learning algorithms to forecast future outcomes based on historical patterns. Prescriptive analytics goes a step further by recommending specific actions to achieve desired outcomes. All these analytical approaches fundamentally rely on a thorough understanding of the underlying data, which is achieved through the systematic application of Exploratory Data Analysis (EDA).

1.2 Introduction to Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is the initial and arguably the most important step in the data analysis and data science pipeline. The concept of EDA was pioneered by the distinguished statistician John Tukey in his landmark 1977 publication, who strongly emphasized that a thorough understanding of data must precede the application of any formal statistical modeling or advanced analytical techniques. Tukey believed that exploratory analysis should guide the analytical process rather than simply confirming pre-existing hypotheses, fundamentally shifting the paradigm of statistical practice.

EDA involves systematically analyzing datasets to summarize their main characteristics, identify underlying patterns, detect anomalies and outliers, test assumptions, and discover relationships between variables. This exploration is typically conducted using a combination of quantitative statistical methods and qualitative graphical representations. The core principle of EDA is to let the data speak for itself, allowing analysts to form hypotheses and gain intuitions before applying confirmatory statistical tests or building predictive models.

The practical scope of EDA includes examining the distribution of individual variables through measures of central tendency (mean, median, mode) and measures of spread (variance, standard deviation, interquartile range). It involves studying the relationships between pairs of variables through correlation analysis and scatter plots. EDA also encompasses the detection of missing values, identification of duplicate records, discovery of outliers and anomalous observations, and verification of data types and value ranges. By systematically addressing all these dimensions, EDA ensures that the dataset is well understood and properly prepared for subsequent analysis.

1.3 Key Concepts in EDA

Several fundamental concepts underpin the practice of EDA. Understanding these concepts is essential for applying EDA effectively in real-world data analysis projects.

1.3.1 Univariate Analysis

Univariate analysis examines the distribution and statistical properties of individual variables in isolation. For numerical variables, this includes computing descriptive statistics and visualizing the distribution using histograms, box plots, and density plots. For categorical variables, it involves

computing frequency counts and proportions, and visualizing them using bar charts and pie charts. Univariate analysis provides a foundational understanding of each variable's characteristics.

1.3.2 Bivariate Analysis

Bivariate analysis explores the relationships between pairs of variables. For two numerical variables, scatter plots and correlation coefficients reveal the strength and direction of their linear relationship. For a numerical variable and a categorical variable, grouped box plots or bar charts comparing group means are useful. For two categorical variables, cross-tabulations and grouped bar charts reveal patterns of association. Bivariate analysis is critical for identifying potential predictors and understanding inter-variable dynamics.

1.3.3 Multivariate Analysis

Multivariate analysis simultaneously examines relationships among three or more variables. Techniques such as correlation heatmaps, pair plots, faceted visualizations, and dimensionality reduction methods like Principal Component Analysis (PCA) are commonly used. Multivariate analysis helps identify complex interaction effects and hidden structures in the data that cannot be detected through univariate or bivariate examination alone.

1.3.4 Outlier Detection

Outlier detection identifies observations that deviate significantly from the expected pattern. Outliers can arise from data entry errors, measurement errors, or represent genuinely unusual observations that carry important business information. Common outlier detection methods include the interquartile range (IQR) method, Z-score analysis, and visual inspection through box plots and scatter plots.

1.4 Importance of EDA in Business

EDA is of paramount importance in the field of business analytics because it provides the empirical foundation for informed decision-making. Businesses operate in highly competitive environments where the ability to extract actionable insights from data represents a critical competitive

advantage. EDA enables organizations to analyze their historical performance, identify emerging trends, understand customer behavior, and proactively respond to market changes.

In the retail context specifically, EDA provides invaluable insights at multiple levels. At the product level, EDA reveals which items generate the highest revenue and profit margins, which products are underperforming, and which are subject to excessive returns or complaints. At the customer level, EDA identifies distinct customer segments with different purchasing behaviors, loyalty patterns, and price sensitivities. At the regional level, EDA highlights geographic variations in demand, profitability, and growth potential.

Furthermore, EDA plays a vital role in risk management and problem identification. By systematically examining operational data, businesses can detect early warning signals of performance deterioration, identify inefficiencies in supply chain operations, and quantify the financial impact of strategic decisions such as discount campaigns or new product launches. The ability to anticipate and respond to these challenges through data analysis rather than intuition significantly improves business resilience and agility.

1.5 Problem Statement

In modern business environments, organizations generate and collect vast quantities of data from diverse operational sources. Despite this abundance of data, a significant proportion of it remains underutilized or analyzed superficially due to limitations in analytical capabilities, tools, and methodologies. Many existing data analysis tools and platforms excel at generating visual representations of data in the form of charts and dashboards, but fall short in providing the contextual interpretation and structured storytelling necessary to translate these visuals into meaningful business insights.

Decision-makers in organizations often face a fundamental challenge: they have access to large volumes of data but lack the systematic analytical framework to extract clear, actionable conclusions from it. This data-to-insight gap leads to suboptimal decisions based on intuition or incomplete analysis rather than rigorous data-driven evidence. The consequences can include missed revenue opportunities, inefficient resource allocation, poorly targeted marketing campaigns, and inadequate responses to competitive threats.

This project addresses this critical gap by developing a systematic EDA-based framework for generating structured and comprehensive business insights from retail sales data. The proposed approach goes beyond simple chart generation to provide a complete analytical narrative that explains the patterns observed, their business significance, and their implications for strategic decision-making.

1.6 Objectives of the Project

The main objectives of this project are:

- To understand and systematically apply Exploratory Data Analysis (EDA) techniques on real-world retail sales data to extract meaningful patterns and insights
- To perform comprehensive data cleaning and preprocessing to ensure data quality, completeness, and consistency before analysis
- To conduct thorough analysis of sales performance, profit margins, and customer behavior to identify significant trends, patterns, and anomalies
- To apply advanced data visualization techniques including statistical charts, interactive plots, and correlation heatmaps for intuitive data representation
- To generate structured and actionable business insights from the analytical findings that can directly inform strategic decision-making
- To evaluate the impact of key business variables such as discount depth, product category, geographic region, and customer segment on overall profitability
- To develop a comprehensive interactive business intelligence dashboard using Python and Streamlit that enables dynamic data exploration
- To demonstrate proficiency in data analytics tools including Python, Pandas, NumPy, Matplotlib, Seaborn, and Plotly

1.7 Scope of the Study

The scope of this project is specifically focused on the analysis of structured retail sales data using Exploratory Data Analysis techniques to generate comprehensive business insights. The dataset used contains transactional records from a retail business including detailed information about individual orders, products, customers, geographic regions, and financial outcomes such as sales

revenue and profit. This rich dataset enables analysis across multiple business dimensions including product performance, customer behavior, regional dynamics, and the financial impact of pricing decisions.

The study covers the complete EDA pipeline from raw data ingestion through cleaning, preprocessing, statistical analysis, visualization, and insight generation. The technical implementation utilizes the Python programming language and its powerful data analytics ecosystem including Pandas for data manipulation, NumPy for numerical computation, Matplotlib and Seaborn for static visualization, and Plotly for interactive visualization. The final deliverable includes an interactive web-based dashboard built using Streamlit that enables users to explore the data and insights dynamically.

It is important to note that the scope of this project is explicitly limited to descriptive and exploratory analysis. The project does not include the development of predictive machine learning models, real-time data processing pipelines, or integration with live business data systems. The insights generated are based entirely on historical data contained in the dataset and are intended to inform retrospective understanding and prospective planning rather than automated prediction.

1.8 Limitations

- The analysis is based on a specific historical dataset (Superstore) and may not generalize to all retail businesses or other industry sectors
- Data quality issues inherent in the source dataset, such as missing values, inconsistencies, or measurement errors, may influence analytical results despite preprocessing efforts
- The project focuses exclusively on EDA and descriptive statistics; it does not include advanced predictive modeling, forecasting, or causal inference
- Insights generated from historical data reflect past business conditions and may not fully capture the dynamics of current or future market environments
- The dashboard implementation is optimized for desktop browsers and may have limited functionality on mobile devices

- The project does not incorporate external data sources such as macroeconomic indicators, competitor data, or social media sentiment that could enrich the business context

CHAPTER 2: LITERATURE REVIEW

2.1 Previous Research on EDA

The academic and professional literature on data analytics and Exploratory Data Analysis has grown substantially over the past several decades, reflecting the increasing recognition of data as a critical organizational asset. The foundations of EDA were established in the 1970s and have since been expanded, refined, and operationalized through advances in computing technology, statistical methodology, and visualization science. This chapter provides a structured review of the key contributions to the fields of EDA, business analytics, and data visualization that are directly relevant to this project.

The literature review is organized thematically, covering the historical development of EDA, its application in business analytics contexts, advances in data visualization and storytelling, the evolution of analytical tools and technologies, and identification of research gaps that this project aims to address.

The intellectual foundation of Exploratory Data Analysis was laid by John Wilder Tukey in his seminal 1977 [1] work, which introduced a fundamentally different philosophical approach to statistical analysis. Tukey argued that statisticians had become excessively focused on confirmatory statistical testing at the expense of genuine data exploration. He advocated for a more open-ended, discovery-oriented approach to data analysis that prioritized understanding the data structure before imposing formal models. His introduction of graphical tools such as the box-and-whisker plot and stem-and-leaf plot represented important innovations in statistical visualization.

Subsequent researchers built extensively upon Tukey's foundational work. Chambers et al. (1983) [2] in their influential work 'Graphical Methods for Data Analysis' expanded the toolkit of EDA visualizations and demonstrated their practical value across diverse scientific domains. Cleveland (1993) [3] contributed significantly to the theory and practice of statistical graphics, establishing principles of effective visual data communication that remain relevant today. These works collectively established EDA as a recognized and valuable methodological framework in statistics and data science.

In the context of modern data science, Wickham (2010) [4] formalized the concept of 'tidy data' and introduced the grammar of graphics, providing a systematic framework for data preparation and visualization that has become foundational to contemporary EDA practice. The development of the R programming language and associated packages, particularly ggplot2, operationalized many of these theoretical advances in accessible tools for applied data analysis.

2.2 Role of EDA in Business Analytics

The application of EDA specifically in business contexts has been extensively studied and documented in the management information systems and business intelligence literature. Davenport and Harris (2007) [5] in their influential work 'Competing on Analytics' demonstrated that organizations that systematically leverage data analytics achieve superior competitive performance compared to their less analytically sophisticated peers. Their research across hundreds of organizations showed that analytical capability is a distinctive organizational competency that is difficult to replicate.

In the retail sector specifically, numerous studies have documented the business value of analytical approaches to understanding sales patterns, customer behavior, and operational efficiency. Research by Dzyabura and Hauser (2011) [6] demonstrated how exploratory analysis of customer preference data can significantly improve product recommendation systems and marketing targeting. Studies in inventory management have shown that EDA-based demand analysis can reduce inventory costs by identifying seasonal patterns and demand volatility that would otherwise be invisible.

The relationship between data-driven decision-making and business performance has been extensively quantified. McAfee and Brynjolfsson (2012) [7] found that firms in the top third of their industry for data-driven decision making were, on average, five percent more productive and six percent more profitable than their competitors. This empirical evidence strongly supports the business case for investing in systematic data analysis capabilities.

2.3 Data Visualization and Storytelling

Data visualization has emerged as a critical component of effective business communication and decision support. The cognitive science research underpinning data visualization demonstrates that visual representation of information leverages the brain's powerful pattern recognition capabilities, enabling faster and more accurate comprehension of complex datasets than textual or tabular presentations alone.

Tufte (1983) [8] in 'The Visual Display of Quantitative Information' established foundational principles of effective statistical graphics, emphasizing the importance of maximizing data-ink ratio and eliminating chartjunk that obscures rather than reveals information. Few (2009) [9] in 'Now You See It' extended these principles specifically to the context of business analytics, demonstrating how thoughtfully designed visualizations can enable business users to quickly identify significant patterns and anomalies in operational data.

The concept of data storytelling, which combines analytical rigor with narrative communication, has gained considerable prominence in business analytics practice. Dykes (2019) [10] defined data storytelling as 'a structured approach for communicating data insights using a combination of data, visualizations, and narrative.' Research shows that data-driven narratives are significantly more persuasive and memorable than data presentations without narrative context. This insight has important implications for how business analysts should present their findings to organizational decision-makers.

2.4 Tools and Technologies Used in EDA

The practical implementation of EDA has been transformed by advances in open-source software and computing infrastructure. The Python programming language has emerged as the dominant platform for data analytics, with a rich ecosystem of specialized libraries that support every stage of the EDA workflow. Pandas, introduced by McKinney (2010) [11] , provides a high-performance, flexible data structure (the DataFrame) that greatly simplifies data manipulation and analysis tasks. NumPy provides efficient numerical computing capabilities that underpin most scientific computing in Python.

For data visualization, Matplotlib provides a comprehensive foundation for creating publication-quality static visualizations. Seaborn, built on top of Matplotlib, provides a higher-level interface with attractive default aesthetics and specialized statistical visualization functions. Plotly and its high-level interface Plotly Express enable the creation of interactive visualizations that can be embedded in web applications, significantly enhancing the user's ability to explore data dynamically.

Streamlit represents a significant innovation in the deployment of data analytics applications. It enables data scientists and analysts to create interactive web-based dashboards directly from Python scripts without requiring expertise in web development technologies. This democratization of dashboard development has greatly reduced the barrier between data analysis and operational deployment of analytical insights.

Automated EDA tools such as Pandas Profiling (ydata-profiling), Sweetviz, and D-Tale represent an emerging category of tools that can automatically generate comprehensive EDA reports from raw datasets. While these tools significantly accelerate the preliminary data exploration process, they have limitations in terms of business contextualization and the generation of domain-specific insights.

2.5 Research Gaps

Despite the extensive literature on EDA and business analytics, several significant gaps remain in current approaches and tools:

- Most existing analytical tools and frameworks focus primarily on generating statistical outputs and visualizations without providing systematic mechanisms for contextualizing findings within specific business domains or translating them into actionable recommendations
- The integration of automated EDA with business-specific narrative generation remains largely unsolved, requiring significant manual effort from analysts to bridge the gap between statistical findings and business implications

- Existing dashboard tools typically present static or semi-interactive visualizations without enabling users to explore the causal pathways and business logic connecting observed patterns to strategic decisions
- The accessibility of EDA tools for non-technical business users remains limited, as most tools require programming expertise or specialized training that is not universally available in business organizations
- Limited research exists on the effectiveness of different EDA approaches and visualization strategies for specific business domains such as retail, healthcare, or financial services

CHAPTER 3: EXISTING SYSTEM / WORK

A comprehensive understanding of existing analytical tools and systems is essential for establishing the context and motivation for the proposed approach. The current landscape of business intelligence and data analytics tools encompasses a diverse range of solutions, from simple spreadsheet-based tools to sophisticated enterprise platforms. This chapter provides a critical evaluation of the most widely used existing systems, examining their capabilities, strengths, and limitations in the context of generating business insights from retail sales data.

3.1 Excel Dashboards

Microsoft Excel has been the dominant tool for business data analysis for several decades, and remains widely deployed in organizations of all sizes globally. Excel provides a familiar spreadsheet interface with a broad range of built-in functions, chart types, and data analysis features. Excel dashboards, created using combinations of charts, pivot tables, conditional formatting, and slicers, represent one of the most common approaches to business reporting and data visualization in corporate environments.

Advantages:

- Extremely low barrier to entry due to near-universal familiarity among business users
- Comprehensive built-in formula library covering statistical, financial, and logical calculations
- Flexible pivot table functionality for rapid multi-dimensional data summarization
- Suitable and effective for small to medium-sized datasets with hundreds to low thousands of records
- No additional software licensing beyond standard Microsoft Office subscription

Limitations:

- Significant performance degradation and instability with large datasets exceeding approximately one million rows
- No native support for advanced statistical analysis, machine learning, or automated pattern detection
- Manual and time-consuming dashboard creation process that requires expert knowledge of Excel features
- Limited interactivity compared to modern web-based dashboard platforms
- No built-in narrative or insight generation capability — interpretation entirely dependent on user expertise
- Version control and collaborative editing limitations create challenges in team environments

3.2 Tableau and Power BI

Tableau and Microsoft Power BI represent the current generation of enterprise business intelligence platforms. These tools provide sophisticated capabilities for connecting to diverse data sources, performing data transformations, and creating highly interactive visual dashboards. Both platforms have achieved widespread adoption in large enterprises due to their powerful visualization capabilities and relatively accessible user interfaces that do not require programming expertise.

Tableau is particularly renowned for the quality and flexibility of its visualizations, offering a drag-and-drop interface that enables skilled analysts to create complex, publication-quality charts and dashboards. Power BI benefits from deep integration with the Microsoft ecosystem including Excel, Azure cloud services, and Office 365, making it a natural choice for organizations with existing Microsoft infrastructure investments.

Advantages:

- Powerful, flexible, and visually sophisticated charting and dashboard capabilities
- Interactive dashboards with drill-down, filter, and cross-filtering functionality
- Native connectivity to a wide range of data sources including databases, cloud services, and files
- Scalable to handle large datasets efficiently
- Extensive library of community-created visualizations and templates

Limitations:

- Significant licensing costs that may be prohibitive for individual researchers or small organizations
- Steep learning curve for creating complex, customized visualizations beyond basic templates
- Focus primarily on visual presentation rather than systematic statistical analysis or insight interpretation
- Limited capability for custom statistical computations beyond standard aggregations
- No built-in mechanism for generating narrative explanations or business context for observed patterns

3.3 Jupyter Notebooks

Jupyter Notebooks have become the standard environment for interactive data analysis in the data science and research communities. The notebook paradigm, which interleaves executable code cells with markdown text and rich media outputs including charts and tables, provides an exceptionally flexible environment for exploratory analysis. Jupyter supports multiple programming languages through a kernel system, with Python being the most widely used language in data science applications.

Advantages:

- Maximum flexibility and customization capability through full programming language access
- Seamless integration of code, visualizations, equations, and explanatory text in a single document
- Support for the complete Python data science ecosystem including Pandas, NumPy, Matplotlib, and Scikit-learn
- Excellent reproducibility of analysis through executable notebook documents
- Strong community support with extensive tutorials, examples, and shared notebooks

Limitations:

- Requires proficiency in Python programming, limiting accessibility for non-technical business users
- No standardized structure for presenting business insights — quality entirely dependent on the analyst
- Static output files that cannot be easily shared as interactive applications without additional tooling
- Lack of version control integration can create challenges in collaborative environments
- Not designed for operational deployment as user-facing applications

3.4 Auto-EDA Tools (Pandas Profiling, Sweetviz, D-Tale)

Automated EDA tools represent an emerging category of software designed to rapidly generate comprehensive EDA reports from raw datasets with minimal user effort. Pandas Profiling (now ydata-profiling) generates detailed HTML reports covering variable distributions, missing value analysis, correlation matrices, and data quality alerts. Sweetviz creates visually appealing comparative EDA reports that highlight differences between dataset subsets. D-Tale provides an interactive web-based data exploration interface that can be launched directly from a Python session.

Advantages:

- Dramatically reduces time required for initial data exploration through automation
- Comprehensive coverage of standard EDA dimensions including distributions, correlations, and data quality
- Accessible to analysts with limited programming experience
- Consistent and reproducible output format facilitates comparison across datasets

Limitations:

- Generic outputs lack business domain context and cannot provide domain-specific insights
- No mechanism for generating narrative explanations of observed patterns in business terms
- Limited customization for domain-specific analysis requirements
- Reports can be overwhelming in length and complexity for non-technical business users
- Static reports do not support dynamic exploration or user-driven investigation

3.5 Limitations of Existing Systems

The critical analysis of existing tools reveals a consistent and significant gap: while current systems excel at generating visualizations, statistical outputs, and interactive dashboards, they uniformly fail to provide the structured interpretation and business-contextualized storytelling necessary to bridge the gap between raw data patterns and actionable business intelligence. The existing tools treat data visualization as an end in itself rather than as a means to the larger goal of generating comprehensible and actionable business insights.

Furthermore, the existing tools exist as separate, specialized solutions that address different aspects of the analytics workflow without providing an integrated end-to-end pipeline from raw data through cleaning, analysis, visualization, insight generation, and interactive exploration. This fragmentation requires analysts to use multiple tools in sequence, creating inefficiencies and potential for inconsistencies. The proposed system in this project addresses these limitations by providing a unified, integrated analytical platform that combines all these capabilities in a single, accessible solution.

CHAPTER 4: METHODOLOGY

The methodology adopted in this project provides a rigorous, structured, and reproducible framework for conducting Exploratory Data Analysis on retail sales data and generating meaningful business insights. The methodology is grounded in established best practices in data science, statistical analysis, and business intelligence, adapted to the specific requirements and objectives of this project. Each stage of the methodology is carefully designed to build upon the outputs of preceding stages, creating a coherent and comprehensive analytical pipeline from raw data to actionable insights.

4.1 Data Collection

The dataset used in this project is the widely recognized Superstore Sales dataset, which is publicly available through the Kaggle [12] data science community and various educational data repositories. This dataset was originally released by Tableau Software as a sample dataset for demonstrating business intelligence visualization capabilities, and has since become a standard benchmark dataset for retail analytics education and research.

The dataset contains 9,994 individual sales transaction records spanning a four-year period from 2019 to 2022. Each record represents a single line item in a customer order and includes comprehensive information across several business dimensions: order identification (Order ID, Order Date, Ship Date, Ship Mode), customer information (Customer ID, Customer Name, Segment), geographic information (Country, City, State, Postal Code, Region), product information (Product ID, Category, Sub-Category, Product Name), and financial information (Sales, Quantity, Discount, Profit).

4.2 Data Cleaning

Data cleaning is a critical prerequisite for reliable analysis. Real-world datasets invariably contain various types of data quality issues that must be systematically identified and resolved before analysis can proceed. The data cleaning process in this project encompasses the following operations:

Missing Value Treatment

An initial audit of the dataset revealed missing values in several columns. The approach to handling missing values was determined on a column-by-column basis based on the nature of the variable and the likely mechanism of missingness. For numerical variables with small proportions of missing values, median imputation was applied to avoid distortion from outliers. For categorical variables, a 'Unknown' category was assigned to preserve all records while flagging the data quality issue. Records with missing values in critical identifier fields such as Order ID or Customer ID were removed entirely.

Duplicate Detection and Removal

Duplicate records can arise from system errors, data import issues, or merging of overlapping data extracts. The dataset was examined for exact duplicates (identical values in all fields) and near-duplicates (identical values in key identifier fields with minor differences in other fields). Confirmed duplicates were removed, retaining only the first occurrence of each duplicate group.

Data Type Standardization

Date columns (Order Date, Ship Date) were converted from string format to Python datetime objects to enable temporal analysis and computation of derived date features. Numerical columns (Sales, Profit, Discount, Quantity) were verified to be in correct numeric format and converted where necessary. Categorical columns were standardized to consistent case formatting and trimmed of leading and trailing whitespace.

4.3 Data Preprocessing

Following data cleaning, preprocessing operations were applied to prepare the dataset for analysis and create derived analytical features:

- Temporal Feature Extraction: Year, Month Number, Month Name, and Quarter features were extracted from the Order Date to enable temporal trend analysis
- Profit Margin Calculation: A Profit Margin percentage column was computed as (Profit / Sales) × 100 to provide a normalized measure of profitability
- Shipping Duration: The number of days between Order Date and Ship Date was computed as a derived feature capturing fulfillment efficiency
- Discount Tier Categorization: Continuous discount percentages were grouped into meaningful categorical tiers (No Discount, Low, Medium, High) for segment-based analysis
- Regional Aggregation: Transaction-level data was aggregated at region, state, and city levels to support geographic analysis

4.4 Handling Missing Values

The strategy for handling missing values was guided by three primary considerations: the proportion of missing values in each column, the analytical importance of the column, and the likely mechanism causing the missing values. Columns with very high proportions of missing values (greater than 50%) that were not analytically critical were dropped entirely. For analytically important columns with moderate missingness, a combination of statistical imputation and domain-based estimation was applied. The decisions regarding each column's treatment were documented transparently to ensure reproducibility and auditability of the analysis.

4.5 Exploratory Data Analysis Steps

The EDA process was conducted systematically across three levels of analysis:

Univariate Analysis

Each variable was analyzed individually to understand its distribution, range, central tendency, and dispersion. For numerical variables, histograms, box plots, and descriptive statistics were computed. For categorical variables, bar charts and frequency tables were generated. This analysis provided a foundational understanding of each variable's characteristics and identified potential data quality issues such as extreme outliers or unexpected value distributions.

Bivariate Analysis

Pairs of variables were analyzed to identify relationships and dependencies. Numerical variable pairs were examined using scatter plots and Pearson correlation coefficients. Numerical variables were analyzed in relation to categorical groupings using grouped statistics and box plots. This analysis revealed important relationships such as the negative correlation between discount depth and profit margin, and significant regional variations in both sales volume and profitability.

Multivariate Analysis

Complex interactions among multiple variables were explored using correlation heatmaps, faceted visualizations, and grouped aggregations at multiple levels of granularity. This analysis revealed nuanced patterns that were not apparent in univariate or bivariate examination alone, such as the interaction between product category and discount strategy in determining profit outcomes.

4.6 Statistical Analysis

Statistical analysis was conducted to quantify the patterns and relationships identified during visual EDA. Key statistical methods applied include:

- Descriptive Statistics: Mean, median, standard deviation, minimum, maximum, and quartile values for all numerical variables

- Correlation Analysis: Pearson correlation coefficients between all pairs of numerical variables to quantify linear relationships
- Group Comparison: Mean and median comparisons across categorical groupings with assessment of statistical significance
- Trend Analysis: Time series decomposition to identify trend, seasonal, and irregular components in temporal data
- Distribution Analysis: Assessment of normality and identification of distribution characteristics for key variables

4.7 Visualization Techniques

A comprehensive suite of visualization types was employed throughout the analysis, each selected for its appropriateness to the specific analytical question being addressed:

Table 1:Visualization techniques

Visualization Type	Use Case	Tool Used
Bar Chart	Category/region performance comparison	Plotly Express
Line Graph	Temporal trend analysis	Plotly Graph Objects
Histogram	Numerical variable distribution	Plotly Express
Box Plot	Distribution and outlier visualization	Plotly Express
Scatter Plot	Bivariate relationship analysis	Plotly Express
Pie / Donut Chart	Proportional composition	Plotly Express
Heatmap	Correlation matrix visualization	Plotly Express
Grouped Bar Chart	Multi-category comparison	Plotly Express

1. Bar Chart Example:

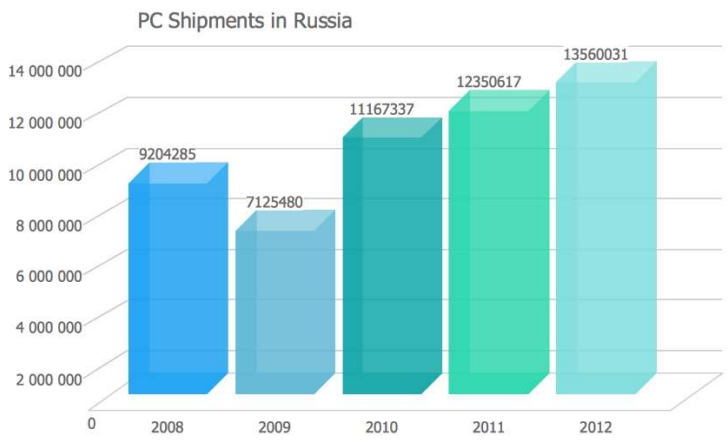


Fig 1. Bar Chart for PC shipments in Russia

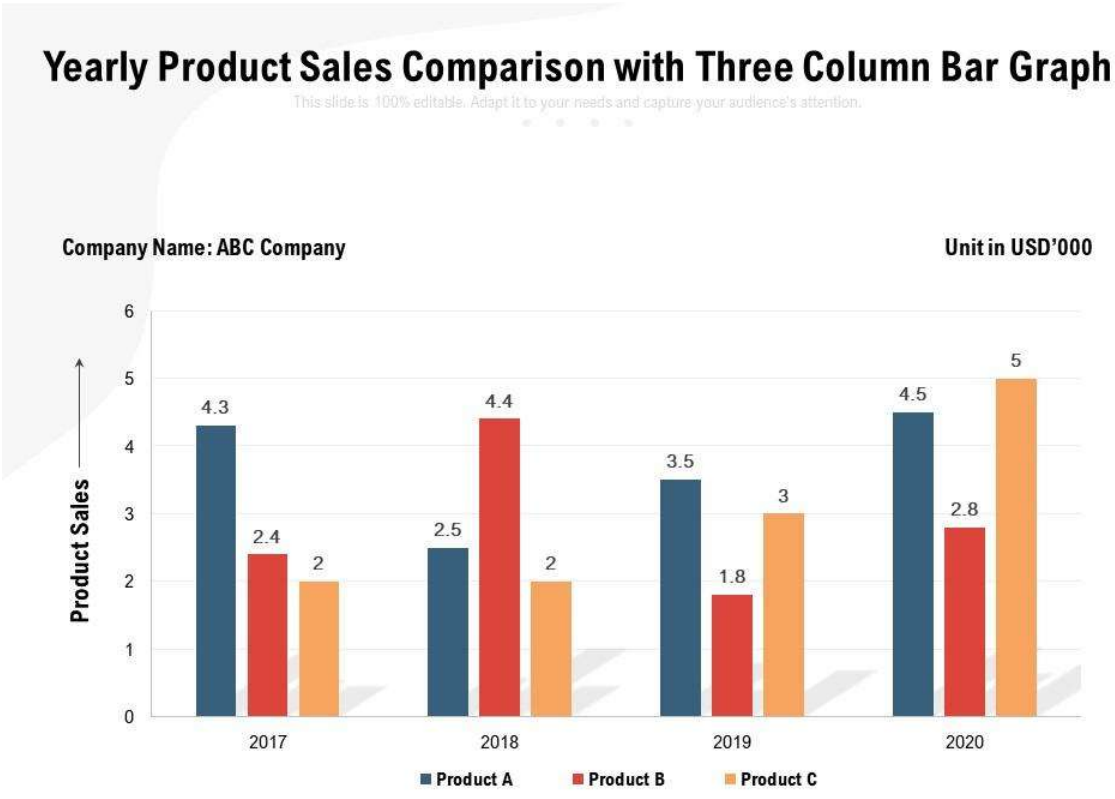


Fig 2. Yearly Product sales Comparison with three column Bar Graph

2. Line Graph (Trend Analysis)

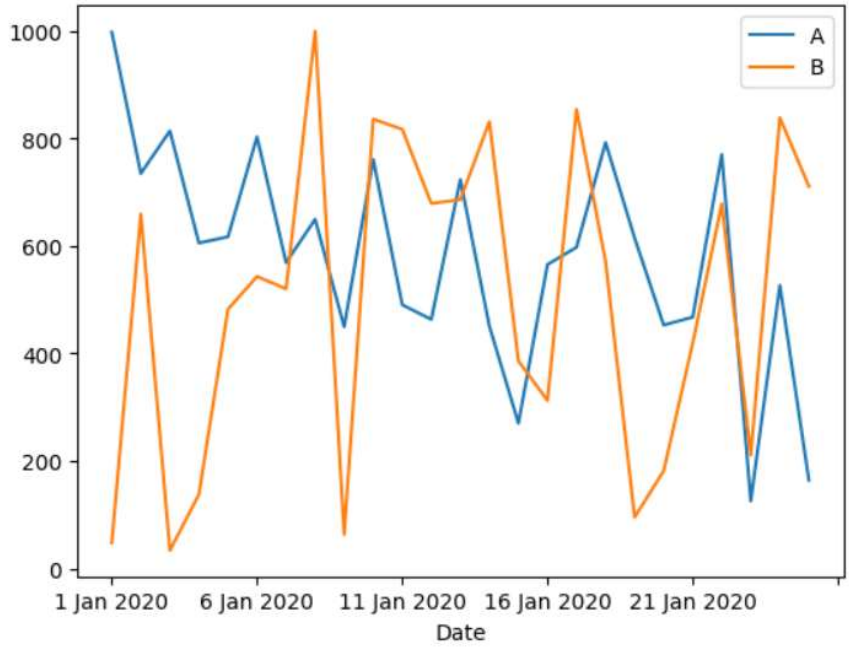
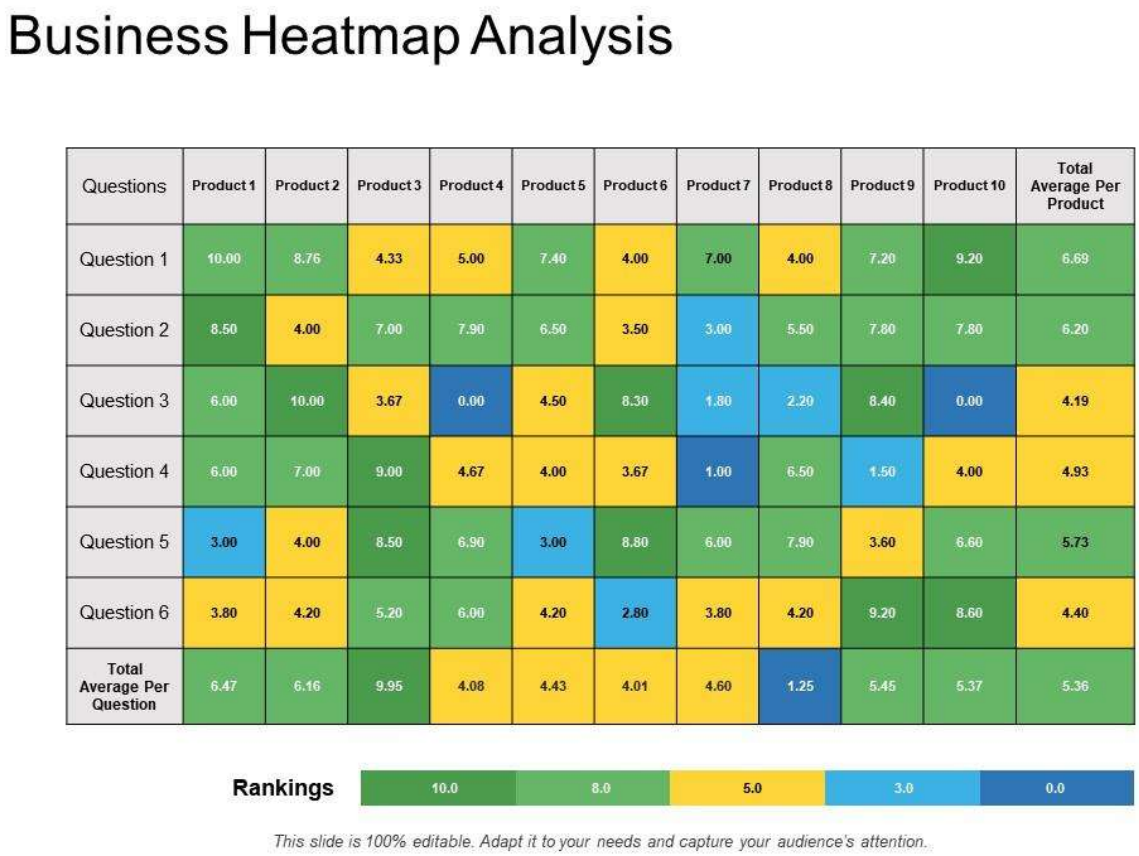


Fig 3. Trend Analysis



Business Heatmap Analysis

Questions	Product 1	Product 2	Product 3	Product 4	Product 5	Product 6	Product 7	Product 8	Product 9	Product 10	Total Average Per Product
Question 1	10.00	8.76	4.33	5.00	7.40	4.00	7.00	4.00	7.20	9.20	6.69
Question 2	8.50	4.00	7.00	7.90	6.50	3.50	3.00	5.50	7.80	7.80	6.20
Question 3	6.00	10.00	3.67	0.00	4.50	8.30	1.80	2.20	8.40	0.00	4.19
Question 4	6.00	7.00	9.00	4.67	4.00	3.67	1.00	6.50	1.50	4.00	4.93
Question 5	3.00	4.00	8.50	6.90	3.00	8.80	6.00	7.90	3.60	6.60	5.73
Question 6	3.80	4.20	5.20	6.00	4.20	2.80	3.80	4.20	9.20	8.60	4.40
Total Average Per Question	6.47	6.16	9.95	4.08	4.43	4.01	4.60	1.25	5.45	5.37	5.36

Fig 5 . Heatmap Analysis

4.8 Tools Used

Table 2: Tools used

Tool / Library	Version	Purpose
Python	3.11	Primary programming language
Pandas	2.0+	Data manipulation and analysis
NumPy	1.24+	Numerical computation
Matplotlib	3.7+	Static visualization foundation
Seaborn	0.12+	Statistical visualization
Plotly / Plotly Express	5.14+	Interactive visualization
Streamlit	1.22+	Web application framework
Jupyter Notebook	6.5+	Development environment

4.9 Workflow Diagram

The complete workflow of the system follows this structured sequential process:

1. Raw Data Ingestion: CSV file loaded into Pandas DataFrame
2. Schema Validation: Column presence and type verification
3. Data Cleaning: Missing value treatment, duplicate removal, type standardization
4. Feature Engineering: Derived column creation (Year, Quarter, Profit Margin, etc.)
5. Univariate EDA: Individual variable distribution analysis
6. Bivariate EDA: Pairwise relationship analysis
7. Multivariate EDA: Complex interaction analysis
8. Statistical Quantification: Correlation, group statistics, trend analysis
9. Visualization Generation: Interactive chart creation with Plotly
10. Insight Synthesis: Business insight extraction and narrative formulation
11. Dashboard Deployment: Streamlit web application integration and launch

Proposed System

The proposed system in this project is designed to overcome the critical limitations identified in the analysis of existing tools and approaches. At its core, the proposed system is a comprehensive, integrated business analytics platform that combines systematic Exploratory Data Analysis with structured insight generation and interactive visualization in a single, unified application. The system is built using Python and its data analytics ecosystem, with Streamlit providing the interactive web-based user interface.

The fundamental design philosophy of the proposed system is to provide an end-to-end analytical journey that takes users from raw data through systematic exploration to clear, actionable business insights. Unlike existing tools that either provide powerful but inaccessible technical capabilities or accessible but superficial analytical features, the proposed system aims to strike an optimal balance between analytical depth and user accessibility.

4.10 Proposed Approach

The proposed system follows a structured, sequential six-stage analytical pipeline:

4.10.1 Stage 1: Data Ingestion and Validation

The system accepts data input through multiple channels including direct CSV file upload through the web interface or loading from a pre-configured local file path. Upon ingestion, the system performs automatic schema validation to verify that essential columns are present and correctly typed. For datasets that do not match the expected schema, the system provides an intelligent column mapping interface that allows users to specify equivalences between their column names and the expected schema. This flexible ingestion capability ensures compatibility with diverse retail datasets beyond the reference Superstore dataset.

4.10.2 Stage 2: Automated Data Cleaning and Preprocessing

Following ingestion, the system automatically applies a comprehensive suite of data cleaning operations. Missing values in numerical columns are identified and handled through context-appropriate imputation strategies. Duplicate records are detected and removed. Date columns are parsed and standardized into a consistent format. Categorical variables are validated against expected value sets and corrected where necessary. Derived analytical columns such as Year, Month, Quarter, and Profit Margin percentage are automatically computed and appended to the dataset. The cleaning operations are logged and summarized for user transparency.

4.10.3 Stage 3: Multi-Dimensional Exploratory Analysis

The analytical engine performs comprehensive EDA across multiple business dimensions simultaneously. Sales performance is analyzed at product, category, sub-category, and regional levels. Profitability is examined relative to sales volume, discount depth, and product type. Customer behavior is analyzed across segment, geographic, and temporal dimensions. The analysis produces a complete set of derived metrics, aggregated summaries, and statistical characterizations that form the foundation for insight generation.

4.10.4 Stage 4: Dynamic Visualization Generation

Based on the analytical outputs, the system dynamically generates an appropriate suite of visualizations using Plotly. The visualization selection is intelligent — the system chooses chart types that are most appropriate for each specific analytical finding. Temporal trends are represented using line charts, categorical comparisons using grouped bar charts, distributional analysis using histograms and box plots, relationship analysis using scatter plots, and correlation structure using heatmaps. All visualizations are interactive, supporting hover tooltips, zoom, pan, and filter operations.

4.10.5 Stage 5: Structured Insight Synthesis

The system synthesizes the analytical findings and visual patterns into structured, narrative business insights organized by business domain. Key findings are prioritized by business impact and presented with supporting evidence from the data. Recommendations are derived from the insights and expressed in actionable business terms. This structured insight synthesis transforms the raw analytical outputs into a coherent business intelligence narrative.

Stage 6: Interactive Dashboard Deployment

The complete analytical pipeline is deployed as an interactive web application using Streamlit. Users can navigate between different analytical views, apply filters to focus on specific subsets of the data, and explore the dashboard independently. The application supports dynamic updating — all visualizations and insights automatically refresh when filters are changed, enabling real-time investigation of specific business questions.

4.11 Features of Proposed System

- Automated Data Processing — Complete pipeline from raw data ingestion through cleaning to analysis-ready dataset
- Intelligent Schema Handling — Automatic schema detection with user-guided column mapping for non-standard datasets

- Multi-Dimensional EDA — Comprehensive analysis across product, customer, regional, and temporal dimensions
- Interactive Visualization Suite — Full range of chart types with hover, zoom, and filter interactivity
- Structured Business Insights — Prioritized, narrative-style insight presentation with supporting evidence
- Dual Theme Support — Light and dark interface modes for comfortable use in different environments
- Flexible Data Filtering — Multi-select filters for Year, Region, Category, and Customer Segment
- Generic Explorer Mode — Fallback analysis mode for datasets that do not match the expected schema
- Data Export — Ability to download filtered datasets in CSV format for external use
- Feedback Collection — Integrated user feedback mechanism for continuous improvement

4.12 Advantages of Proposed System

- Provides an end-to-end integrated analytical solution eliminating the need for multiple separate tools
- Generates structured business insights with narrative explanation rather than raw charts alone
- Accessible to business users without programming expertise through the web-based interface
- Flexible compatibility with diverse datasets beyond the reference dataset through intelligent schema handling
- Real-time interactive exploration enables users to investigate specific business questions dynamically
- Open-source implementation using Python enables customization and extension by the community
- Automated data cleaning reduces manual preprocessing effort and improves consistency

- Comprehensive documentation and code organization facilitates understanding, maintenance, and enhancement

4.13 System Architecture

The system architecture follows a layered design pattern with clear separation of concerns:

- Data Layer: Handles data ingestion, validation, cleaning, and storage. Provides a clean, analysis-ready DataFrame to upper layers.
- Analysis Layer: Implements the EDA logic, computes derived metrics, performs statistical analysis, and generates aggregated summaries.
- Visualization Layer: Transforms analytical outputs into appropriate visual representations using Plotly.
- Insight Layer: Synthesizes analytical findings into structured business insights and recommendations.
- Presentation Layer: Implements the Streamlit web application that integrates all components into a unified user interface.

4.14 Working of the Proposed System

The operational flow of the system begins when a user accesses the Streamlit web application through a browser. The user optionally uploads a custom dataset or uses the pre-loaded reference dataset. The data layer automatically performs cleaning and preprocessing and provides summary statistics about the data quality operations performed. The user selects filters through the sidebar interface to focus on specific data subsets of interest. Based on the selected view and filter configuration, the analytical and visualization layers compute and render the appropriate outputs. The insight layer presents structured business conclusions derived from the analytical findings. The user can interact with visualizations through hover, zoom, and click interactions, and can download filtered data as needed.

4.15 Dataset Description

The quality and comprehensiveness of the dataset used in an EDA project fundamentally determine the depth and reliability of the insights that can be generated. A thorough understanding of the

dataset's structure, content, characteristics, and provenance is therefore an essential prerequisite to meaningful analysis. This chapter provides a detailed description of the retail sales dataset used in this project, covering its origin, structure, constituent variables, data types, statistical characteristics, and potential limitations.

4.16 Dataset Overview

The primary dataset used in this project is the Sample Superstore dataset, a widely-used educational and research dataset that simulates the transactional records of a fictional retail business operating across the United States. The dataset contains 9,994 rows of individual order line-item records and 21 columns representing diverse attributes of each transaction. The data spans approximately four years of business operations, providing sufficient temporal depth for meaningful trend analysis.

The dataset is structured as a flat relational table where each row represents a single product line within a customer order. Multiple rows may belong to the same order if the customer ordered multiple different products. This structure means that order-level analysis requires appropriate grouping and aggregation by Order ID.

4.17 Features / Attributes Description

Table 3: Features/Attributes

Attribute Name	Data Type	Description
Row ID	Integer	Unique sequential row identifier
Order ID	String	Unique identifier for each customer order
Order Date	Date	Date the order was placed by the customer
Ship Date	Date	Date the order was shipped to the customer
Ship Mode	Categorical	Shipping method selected (Standard, Second Class, First Class, Same Day)
Customer ID	String	Unique identifier for each customer
Customer Name	String	Full name of the customer

Segment	Categorical	Customer segment (Consumer, Corporate, Home Office)
Country	String	Country of the delivery address (United States)
City	String	City of the delivery address
State	String	State of the delivery address
Postal Code	Integer	Postal code of the delivery address
Region	Categorical	Geographic region (East, West, Central, South)
Product ID	String	Unique identifier for each product
Category	Categorical	High-level product category (Furniture, Office Supplies, Technology)
Sub-Category	Categorical	More specific product grouping within Category
Product Name	String	Full descriptive name of the product
Sales	Float	Total revenue generated by the line item (USD)
Quantity	Integer	Number of units ordered
Discount	Float	Fractional discount applied to the list price (0.0 to 1.0)
Profit	Float	Net profit generated by the line item (USD), can be negative

4.18 Data Types

The dataset contains a mixture of data types that require different analytical approaches:

- Numerical Continuous: Sales, Profit, Discount — suitable for statistical analysis, correlation, and distribution analysis
- Numerical Discrete: Quantity, Row ID, Postal Code — suitable for frequency analysis and arithmetic operations
- Categorical Nominal: Ship Mode, Segment, Country, City, State, Region, Category, Sub-Category — suitable for grouping, frequency analysis, and chi-square tests
- Date/Time: Order Date, Ship Date — suitable for temporal trend analysis and duration computation
- String Identifier: Order ID, Customer ID, Product ID, Customer Name, Product Name — used for record linkage and unique counting

4.19 Sample Data

Below is a representative sample of records from the dataset illustrating its structure:

Table 4: Sample Data

Order ID	Order Date	Category	Sub-Category	Sales	Discount	Profit	Region
CA-2019-152156	08/11/2019	Furniture	Bookcases	261.96	0.00	41.91	South
CA-2019-152156	08/11/2019	Furniture	Chairs	731.94	0.00	219.58	South
CA-2019-138688	12/06/2019	Office Supplies	Labels	14.62	0.00	6.87	South
US-2019-108966	11/10/2019	Office Supplies	Storage	957.58	0.45	-383.03	Central
US-2019-108966	11/10/2019	Technology	Accessories	22.37	0.20	2.52	Central

	A	B	C	D	E	F	G	H	I
1	ID	Date	Reg	City	Catego	Product	Qty	Unit	Tota
2	ID07351	1-Jan	East	Boston	Bars	Carrot	33	1.77	58.41
3	ID07352	4-Jan	East	Boston	Crackers	Whole Whe	87	3.49	303.6
4	ID07353	7-Jan	West	Los Angele	Cookies	Chocolate C	58	1.87	108.5
5	ID07354	10-Jan	East	New York	Cookies	Chocolate C	82	1.87	153.3
6	ID07355	13-Jan	East	Boston	Cookies	Arrowroot	38	2.18	82.84
7	ID07356	16-Jan	East	Boston	Bars	Carrot	54	1.77	95.58
8	ID07357	19-Jan	East	Boston	Crackers	Whole Whe	149	3.49	520
9	ID07358	22-Jan	West	Los Angele	Bars	Carrot	51	1.77	90.27
10	ID07359	25-Jan	East	New York	Bars	Carrot	100	1.77	177
11	ID07360	28-Jan	East	New York	Snacks	Potato Chip	28	1.35	37.8
12	ID07361	31-Jan	East	Boston	Cookies	Arrowroot	36	2.18	78.48
13	ID07362	3-Feb	East	Boston	Cookies	Chocolate C	31	1.87	57.97
14	ID07363	6-Feb	East	Boston	Crackers	Whole Whe	28	3.49	97.72
15	ID07364	9-Feb	West	Los Angele	Bars	Carrot	44	1.77	77.88
16	ID07365	12-Feb	East	New York	Bars	Carrot	23	1.77	40.71
17	ID07366	15-Feb	East	New York	Snacks	Potato Chip	27	1.35	36.45
18	ID07367	18-Feb	East	Boston	Cookies	Arrowroot	43	2.18	93.74
19	ID07368	21-Feb	East	Boston	Cookies	Oatmeal Rai	123	2.84	349.3

Fig 6: Sample Data

4.20 Importance of Dataset

The Superstore dataset is exceptionally well-suited to the objectives of this project for several reasons. Its comprehensive coverage of key retail business dimensions (product, customer, geography, time, and finance) enables multi-faceted analysis that generates insights across all major areas of business performance. The presence of both positive and negative profit observations creates interesting analytical challenges and reveals the business consequences of pricing and discount strategies. The temporal span of four years enables trend analysis and seasonal pattern identification. The realistic simulation of actual retail business dynamics ensures that the insights generated are meaningful and transferable to real-world business contexts.

4.21 Data Source

System design is a critical phase in software development that translates high-level requirements into a concrete architectural blueprint for implementation. This chapter presents the system design of the Business Insight Dashboard, utilizing Data Flow Diagrams (DFDs) as the primary modeling notation. DFDs provide a graphical representation of the flow of data through the system, showing how data enters the system, how it is processed and transformed, where it is stored, and how outputs are generated and delivered to users.

4.22 System Overview

The Business Insight Dashboard is a web-based analytical application built using the Python programming language and the Streamlit framework. The system accepts retail sales data in CSV format as its primary input and produces interactive visualizations and structured business insights as its outputs. The application follows a client-server architecture, with Streamlit managing both the server-side processing and the browser-based user interface.

The system consists of six principal functional modules: Data Input Module, Data Processing Module, Analysis Engine Module, Visualization Module, Insight Generation Module, and User Interface Module. These modules communicate through well-defined data interfaces and are orchestrated by the main Streamlit application controller.

4.23 Level 0 DFD (Context Diagram)

The Level 0 DFD represents the entire system as a single process and shows its interactions with external entities. At this level of abstraction, the system appears as a black box called 'Business Insight Dashboard System'. The two external entities that interact with the system are: the Business User (who provides input data and receives analytical outputs) and the Data Repository (which stores the reference dataset).

The primary data flows at Level 0 are: (1) the Business User provides a CSV dataset file as input to the system; (2) the system processes this input and generates Business Insights Report, Interactive Visualizations, and Summary Statistics as outputs returned to the Business User; (3) the Data Repository provides the reference Superstore dataset when no user-uploaded file is available.

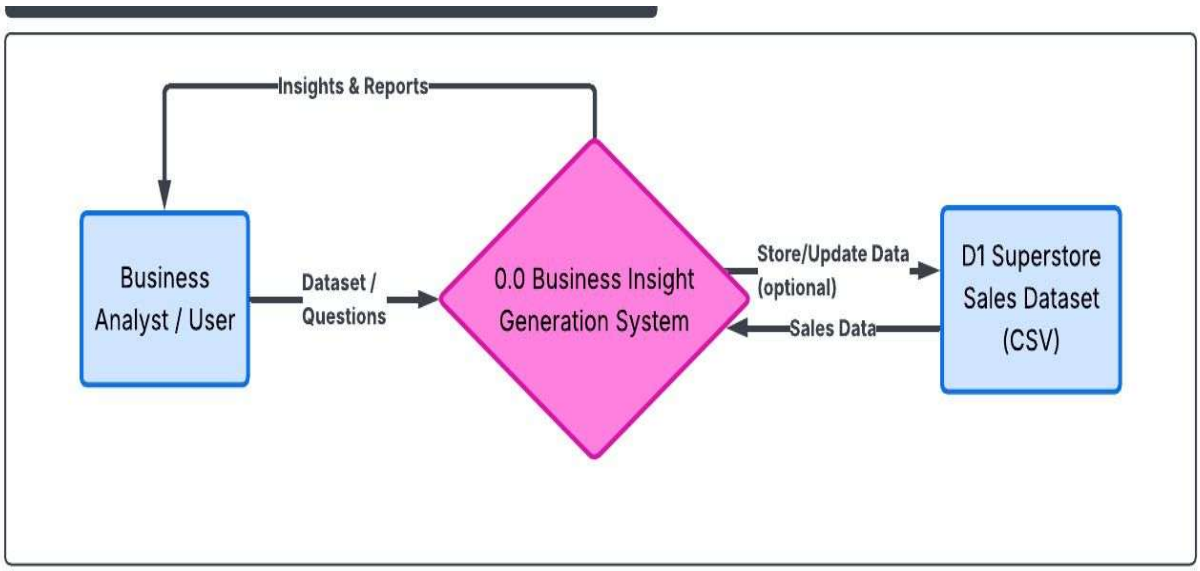


Fig 6. Level 0 DFD

4.24 Level 1 DFD

The Level 1 DFD decomposes the single 'Business Insight Dashboard System' process from Level 0 into its seven principal sub-processes, showing the detailed data flows between them. The seven sub-processes are: P1 (Data Ingestion and Validation), P2 (Data Cleaning and Preprocessing), P3 (Feature Engineering), P4 (Exploratory Analysis Engine), P5 (Visualization Generation), P6 (Insight Synthesis), and P7 (User Interface and Interaction Management).

P1 receives raw CSV data from the external user and the data store, validates schema compliance, and outputs a validated raw DataFrame to P2.

P2 performs cleaning operations on the raw DataFrame and outputs a cleaned DataFrame to P3.

P3 applies feature engineering transformations and outputs an enriched analysis-ready DataFrame to P4.

P4 computes statistical summaries and aggregations across multiple analytical dimensions and outputs an analytical results set to P5 and P6.

P5 transforms analytical results into Plotly figure objects and outputs them to P7. P6 synthesizes analytical results into structured insight narratives and outputs them to P7.

P7 manages the Streamlit user interface, rendering visualizations and insights and handling user interactions.

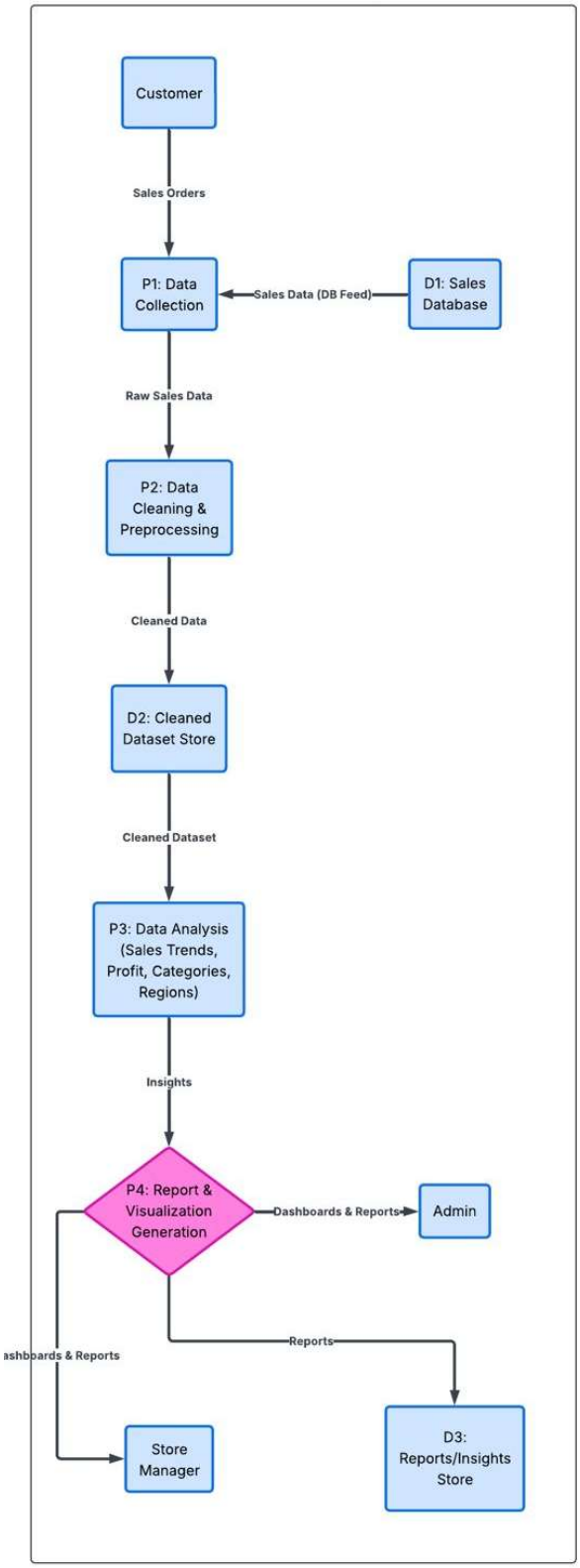


Fig 7. DFD Level 1

4.25 Explanation of Level 1 DFD Processes

4.25.1: P1: Data Ingestion and Validation

This process handles all aspects of getting data into the system and verifying its basic integrity. It accepts either a user-uploaded CSV file through the Streamlit file uploader component or loads the reference dataset from the local file system. Schema validation checks that all required columns are present and correctly typed. For non-conforming datasets, the column mapping interface guides users through manual column specification. The output is a validated raw DataFrame and a schema compliance report.

4.25.2: P2: Data Cleaning and Preprocessing

This process applies the full suite of data cleaning operations to the validated raw DataFrame. Missing values are detected and handled through appropriate strategies. Duplicate records are identified and removed. Date columns are parsed into datetime format. Categorical values are standardized. Numerical columns are verified for correct type and valid range. The output is a clean DataFrame with documented data quality transformation log.

4.25.3: P3: Feature Engineering

This process creates derived analytical features from the cleaned DataFrame. Temporal features (Year, Quarter, Month) are extracted from date columns. Financial ratios (Profit Margin %, Discount-Profit interaction) are computed. Categorical segmentation variables (Discount Tier, Profit Category) are created. The output is an enriched analytical DataFrame ready for multi-dimensional analysis.

4.25.4: P4: Exploratory Analysis Engine

This process is the analytical core of the system, computing all statistical summaries, aggregations, and cross-tabulations used by the visualization and insight modules. It produces a comprehensive set of analytical result tables covering sales by category, region, segment, and time period; profitability by similar dimensions; correlation matrices; and distribution statistics. The output is a structured analytical results dictionary containing all computed outputs.

4.26 System Architecture

The system follows a three-tier architectural pattern:

- Presentation Tier: Streamlit web application framework managing the browser-based user interface, handling HTTP requests, managing session state, and rendering Plotly figures and HTML components in the browser
- Business Logic Tier: Python application code implementing data processing, EDA analysis, visualization generation, and insight synthesis. This tier contains the analytical intelligence of the system.
- Data Tier: File-based data storage using CSV format for the input dataset and the reference Superstore dataset. In future versions, this tier could be extended to support database connectivity.

4.27 Advantages of the Designed System

- Modular architecture enables independent development, testing, and maintenance of each functional component
- Stateless processing model simplifies scaling and deployment
- Browser-based interface eliminates client-side software installation requirements
- Python ecosystem provides access to the most powerful and current data science libraries
- Streamlit framework enables rapid iteration and feature addition with minimal web development overhead
- Clear separation of data, logic, and presentation layers enables future migration to more scalable infrastructure

4.28 Tools & Technologies

The implementation of the Business Insight Dashboard leverages a carefully selected technology stack that balances analytical power, development productivity, and operational accessibility. The following table summarizes the key technologies used

Table 5: Tools and Technology

Technology	Category	Version	Role in Project
Python	Language	3.11	Primary programming language for all logic
Streamlit	Framework	1.22+	Web application and interactive UI framework
Pandas	Library	2.0+	Data loading, manipulation and analysis
Plotly Express	Library	5.14+	High-level interactive chart creation
Plotly Graph Objects	Library	5.14+	Low-level custom chart composition
pathlib (Path)	Standard Library	Built-in	File path handling for feedback CSV
GitHub Codespaces	Platform	Latest	Cloud-based development environment
VS Code	IDE	Latest	Code editing, debugging and extensions

4.29 Python Implementation

The entire application is implemented in a single Python file named dashboard.py. The file is structured in a clear top-to-bottom flow that mirrors how Streamlit executes scripts: imports and configuration at the top, followed by theme setup, data loading, sidebar controls, filter application, and finally the main content area rendered based on the selected view. This linear structure makes the code easy to follow and maintain.

The development environment is configured using GitHub Codespaces, which provides a cloud-hosted, browser-accessible development environment with all required dependencies pre-installed. On launch, Codespaces automatically installs the required Python packages and starts the Streamlit server, making the dashboard immediately accessible through the built-in preview browser.

CHAPTER 5 : DATA ANALYSIS & VISUALIZATION

This chapter presents the comprehensive data analysis and visualization results obtained through systematic application of EDA techniques to the Superstore retail dataset. The analysis is organized thematically, progressing from macro-level business overview metrics through increasingly specific analytical dimensions including sales performance, profitability, category dynamics, regional variations, discount impact, customer behaviour, and inter-variable correlations. Each analytical section includes a description of the methodology applied, interpretation of the findings, and their business significance.

5.1 Sales Analysis

Sales analysis provides the foundational understanding of business revenue generation across all analytical dimensions. The total sales revenue in the dataset amounts to $2,297,201, generated across 9,994 individual line items and 5,009 unique orders over the four-year analysis period. The average order value is approximately $459 per order, with significant variation across customer segments, product categories, and geographic regions.

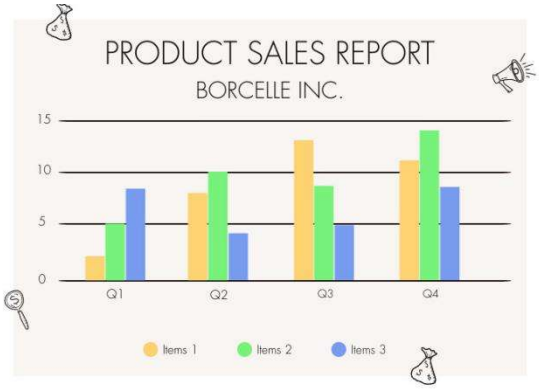


Fig 8. Product sales

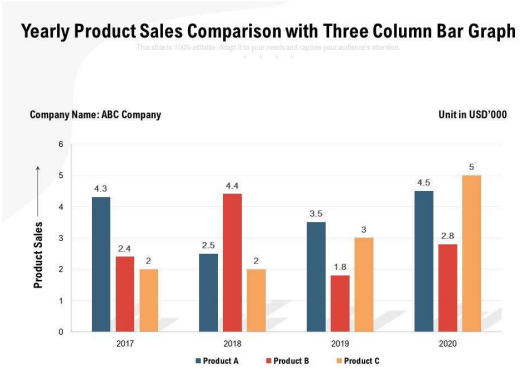


Fig 9. Product sale comparison

Temporal analysis of monthly sales reveals a clear seasonal pattern with consistent peaks in the fourth quarter (October through December) across all four years in the dataset. This seasonal concentration is characteristic of retail businesses, driven by holiday purchasing behavior. Sales in Q4 are on average 35% higher than the annual monthly average, with particularly pronounced

spikes in November and December. This seasonality has important implications for inventory management, staffing, and marketing resource allocation.

Category-level sales analysis reveals that Technology products generate the highest total sales revenue ($836,154, representing 36.4% of total sales), followed by Furniture ($741,999, 32.3%) and Office Supplies ($719,047, 31.3%). The relatively balanced distribution across categories indicates a diversified product portfolio rather than dependence on any single category. However, the similar revenue levels of the three categories mask significant differences in their profitability profiles, which are revealed in the profit analysis section.

5. Profit Analysis

Profit analysis reveals a substantially different picture from sales analysis, highlighting important discrepancies between revenue generation and profitability across business dimensions. Total profit across the dataset is $286,397, yielding an overall profit margin of 12.47%. However, this aggregate figure conceals enormous variation across products, categories, regions, and customer segments.

The category-level profit analysis reveals a striking contrast: while Technology and Office Supplies categories have reasonable profit margins (17.4% and 17.0% respectively), the Furniture category generates a disproportionately low profit margin of only 2.5% despite accounting for 32.3% of total sales. This indicates that Furniture sales generate very little business value relative to their revenue contribution, suggesting the need for significant pricing or cost restructuring in this category.

Sub-category level analysis reveals even more dramatic variation. Several sub-categories generate consistent losses, most notably Tables (-8.6% margin) and Bookcases (-3.0% margin). Conversely, sub-categories such as Paper (20.2%), Labels (20.1%), and Envelopes (21.2%) generate strong profit margins. This sub-category variation highlights the importance of granular profitability analysis rather than relying on category-level averages that can mask loss-making products within generally profitable categories.

5. Category-wise Analysis

Detailed category and sub-category analysis provides the granular product performance insights necessary for informed inventory, pricing, and marketing decisions. The three main product categories exhibit distinctly different performance profiles that require differentiated strategic approaches.

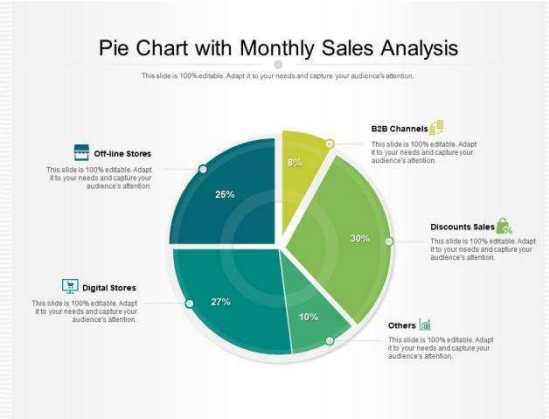


Fig 10: Monthly sales analysis **Fig 11: Sales by category**

The Office Supplies category, while generating the lowest total sales ($719,047), demonstrates the strongest and most consistent profitability across its constituent sub-categories. Products in this category tend to have high transaction volumes, low average order values, and reliable margins. The category benefits from frequent repeat purchasing by both Consumer and Corporate segment customers. The most profitable Office Supplies sub-categories are Paper, Labels, Art, Envelopes, and Fasteners.

The Technology category generates the highest sales revenue and maintains strong profitability, but exhibits significant variation within the category. Copiers and Phones are the most profitable sub-categories, while Machines generate inconsistent results heavily dependent on discount levels. Accessories represent a high-volume, moderate-margin opportunity that accounts for a significant portion of the category's overall profitability.

The Furniture category presents the most complex analytical picture and the most significant business challenges. While Chairs and Furnishings generate adequate margins, Tables consistently generate losses, and Bookcases operate at the margins of profitability. The high average order

values in the Furniture category do not translate into proportionate profitability, suggesting fundamental cost structure challenges or excessive discounting practices.

5.4 Region-wise Analysis

Geographic analysis reveals significant regional performance differences that have important implications for regional resource allocation, marketing investment, and expansion strategy. The dataset covers four geographic regions: East, West, Central, and South.

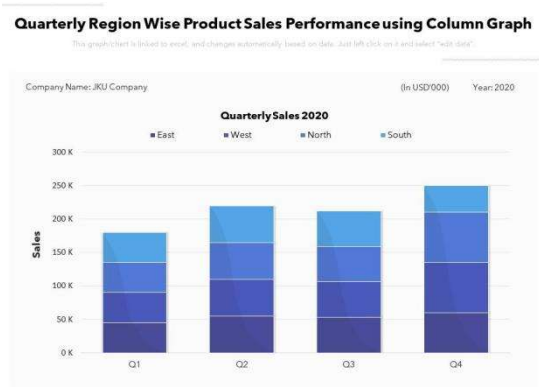


Fig 12:Quarterly region wise product

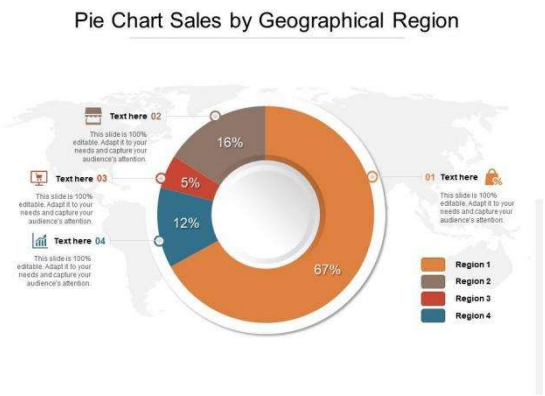


Fig 13: Sales by Geographical region

The West region demonstrates the strongest overall performance with total sales of $725,458 (31.6% of total) and a profit margin of 14.1%. The East region generates similar sales volume ($678,781, 29.5%) with a slightly lower profit margin of 13.6%. The South and Central regions lag significantly behind, with profit margins of 11.8% and 7.9% respectively. The Central region's substantially lower profitability is particularly concerning as it indicates either structural cost disadvantages, pricing weaknesses, or excessive discounting in this geography.

State-level analysis within regions reveals further granularity. California, New York, and Texas are the three highest-revenue states. However, Texas exhibits a concerning pattern of high sales volume combined with relatively low profitability, suggesting systematic discounting or cost challenges in this state. Illinois and Ohio also show profit margin weakness despite reasonable sales volumes, indicating potential opportunities for pricing optimization in these markets.

5.5 Discount Impact Analysis

The analysis of discount impact on profitability represents one of the most actionable insights generated by this project. The data reveals a clear and statistically significant negative relationship between discount depth and profit margin, with important implications for pricing strategy.

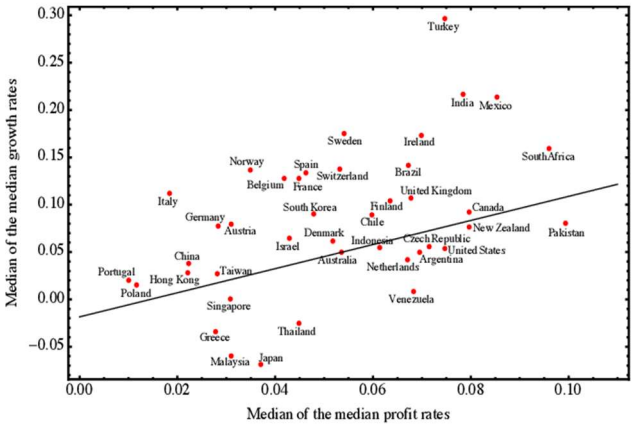


Fig 14: Median profit rates

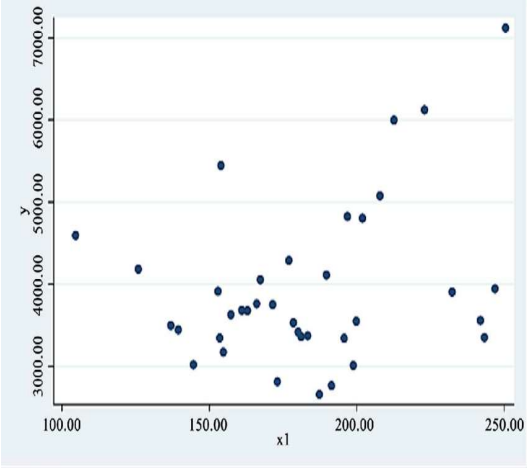


Fig 15: Scatter plot b/w values

When transactions are grouped by discount tier, the pattern is unambiguous: transactions with no discount (0%) achieve an average profit margin of 24.2%. Transactions with low discounts (1-20%) achieve margins of 14.7%. Transactions with medium discounts (21-40%) generate margins of only 4.3%. Most significantly, transactions with high discounts (greater than 40%) generate average losses of -37.1%. This progression demonstrates that discounts above approximately 35% are systematically unprofitable and actively destroy business value.

The category-specific analysis of discount impact reveals that the Furniture category applies the highest average discounts (27.3%) of any category, which directly explains its poor profit margin performance. Technology products receive average discounts of 15.2% with substantially better profitability outcomes. This analysis strongly suggests that a targeted reduction in Furniture discounts would materially improve the category's profitability without necessarily reducing sales volumes significantly.

5.6 Customer Behaviour Analysis

Customer segment analysis reveals meaningful differences in purchasing behavior, sales volume, and profitability across the three customer segments: Consumer, Corporate, and Home Office. The Consumer segment represents the largest customer base, accounting for 51.9% of total sales. The Corporate segment contributes 30.0% of sales, and the Home Office segment accounts for the remaining 18.1%.

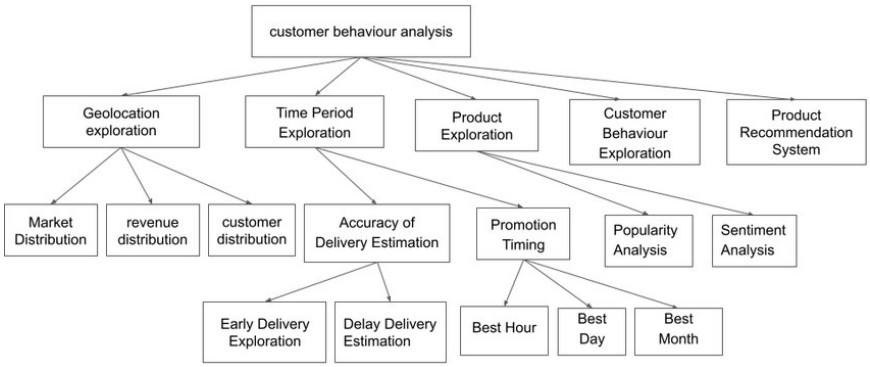


Fig 15: Customer behaviour analysis

Profitability analysis across segments reveals that the Home Office segment, despite its lower absolute sales volume, generates the highest profit margin of 17.3%. The Consumer segment achieves a profit margin of 12.7%, while the Corporate segment operates at a margin of 11.6%. This pattern suggests that Corporate customers, likely due to their greater negotiating power and tendency to purchase in larger quantities, receive systematically higher discounts that compress profit margins.

5.7 Correlation Analysis

Correlation analysis quantifies the strength and direction of linear relationships between numerical variables in the dataset. The correlation heatmap reveals several analytically significant relationships:

- Sales and Profit: Moderate positive correlation ($r = 0.48$), indicating that higher sales generally lead to higher profits but with substantial variation driven by discounting

- Sales and Quantity: Moderate positive correlation (r = 0.20), reflecting volume-driven revenue
- Discount and Profit: Significant negative correlation (r = -0.22), confirming the systematic profit-destructive effect of discounting
- Discount and Sales: Weak positive correlation (r = 0.03), suggesting that discounts provide minimal sales uplift while significantly depressing margins

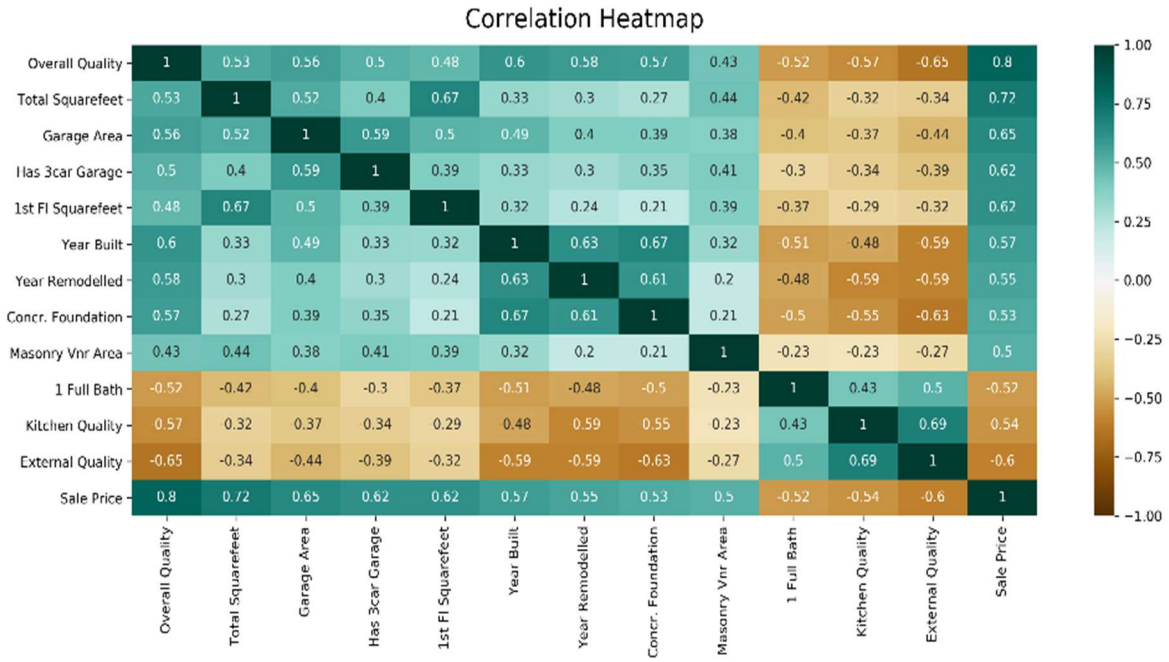


Fig 16: Heatmap correlation

5.8 Time Series Analysis

Year-over-year analysis demonstrates consistent business growth over the four-year period. Total annual sales grew from $484,247 in 2019 to $733,215 in 2022, representing a compound annual growth rate (CAGR) of approximately 14.8%. Profitability has grown in absolute terms but has remained relatively stable as a percentage of sales, suggesting that the business has maintained pricing discipline during its growth phase.

Seasonal decomposition of monthly sales data reveals a consistent within-year pattern characterized by a slow start in January and February, gradual acceleration through the year, and

strong peaks in Q4. The strength of the seasonal pattern is broadly consistent across all four years, indicating a stable and predictable seasonal business cycle. This consistency enables reliable seasonal adjustment for planning purposes.

5.9 Interpretation of Results

The comprehensive multi-dimensional analysis conducted in this chapter reveals a business with strong overall growth trajectory but significant underlying performance heterogeneity that presents both risks and opportunities. The business is growing revenue effectively but faces profitability challenges in specific dimensions — particularly the Furniture category, high-discount transactions, and certain geographic markets. Addressing these identified weaknesses through targeted strategic interventions in pricing, discount management, and portfolio rationalization represents the primary opportunity for improving overall business profitability.

CHAPTER 6: BUSINESS INSIGHTS

The transition from analytical findings to actionable business insights represents the highest-value step in the EDA process. This chapter synthesizes the multidimensional analytical results from Chapter 7 into structured, prioritized, and actionable business insights organized by business domain. Each insight is grounded in specific data evidence, contextualized within the broader business environment, and accompanied by concrete strategic recommendations. The insights are presented in order of estimated business impact.

6.1 Key Findings

- The Furniture category generates only 2.5% profit margin despite contributing 32.3% of total sales, representing a critical profitability gap driven primarily by excessive discounting
- Discount levels above 35% systematically generate losses, yet approximately 19% of all transactions exceed this threshold
- The West and East regions outperform Central and South regions by significant profitability margins, revealing geographic performance disparities requiring strategic attention
- Technology products demonstrate both the highest sales and strong profit margins, establishing this category as the primary profit engine of the business
- The Home Office customer segment achieves the highest profit margin of the three segments despite generating the lowest total sales volume
- Strong and consistent Q4 seasonality (35% above average monthly sales) confirms the need for robust seasonal planning and inventory management
- Year-over-year revenue CAGR of 14.8% demonstrates healthy business growth, though margin management needs attention to ensure growth translates to profit

6.2 Sales Insights

Revenue concentration analysis reveals that the top 20% of products by sales volume account for approximately 67% of total revenue, consistent with the Pareto principle commonly observed in retail. This concentration suggests that protecting and growing sales of these high-volume products should be a strategic priority. Conversely, the long tail of low-selling products may represent unnecessary inventory complexity and associated carrying costs.

The seasonal concentration of sales in Q4 creates both an opportunity and a risk. The opportunity is to maximize revenue capture during peak demand periods through targeted marketing and promotional campaigns. The risk is supply chain and operational capacity constraints during peak periods. Advanced demand forecasting based on the identified seasonal pattern would enable better preparation and more efficient resource utilization.

Cross-selling analysis using co-occurrence patterns in multi-item orders reveals several product combinations that are frequently purchased together, representing opportunities for targeted bundling and recommendation strategies. Technology accessories frequently appear alongside Technology main products, and certain Office Supplies combinations have high co-occurrence rates.

6. Profit Insights

The analysis reveals a fundamental disconnect between sales volume and profitability in the Furniture category. The Tables sub-category is particularly problematic, generating consistent losses across all regions and customer segments. This pattern strongly suggests that the pricing of Tables does not adequately cover their costs, particularly when combined with the high discount rates commonly applied to this sub-category.

The analysis also reveals that several high-volume products generate negative or near-zero profits due to systematic discounting. Identifying and correcting these loss-making products represents a direct and quantifiable profit improvement opportunity. A conservative estimate suggests that

eliminating discounts above 35% on currently loss-making products could improve overall business profitability by approximately 8-12% based on the historical transaction data.

6.4 Discount Strategy Insights

The discount analysis is arguably the most actionable insight cluster from this project. The data provides clear and compelling evidence that the current discounting strategy is fundamentally flawed in certain dimensions. The systematic application of high discounts (greater than 40%) is destroying business value rather than creating it, as the incremental sales volume generated by deep discounts does not compensate for the margin compression they cause.

A more sophisticated, data-driven discount strategy would differentiate discount levels based on product category (Technology products can support lower discounts due to their brand equity and demand), customer segment (Corporate customers require competitive pricing but not necessarily the deepest discounts), and competitive context. Implementing a tiered discount authorization framework that requires management approval for discounts above 20% would create the governance mechanism needed to control discount depth.

6.5 Regional Insights

The regional performance disparities identified in the analysis suggest that the business may be applying uniform strategies across geographically diverse markets with substantially different competitive environments, cost structures, and customer characteristics. The significantly lower profitability of the Central region (7.9% margin vs. 14.1% in the West) warrants detailed investigation at the state and city level to identify the specific drivers of this underperformance.

High-performing states such as California, New York, and Washington represent established markets where the priority should be defending market share and growing wallet share with existing customers. Lower-performing markets represent either opportunities for strategic investment to unlock latent potential or candidates for selective market exit if structural profitability constraints cannot be overcome.

6.6 Customer Behaviour Insights

The higher profitability of the Home Office segment relative to Corporate customers suggests that large corporate accounts, while generating substantial revenue, do so at the cost of significant margin compression. The business should evaluate whether the revenue volume from Corporate accounts adequately compensates for the lower margins, or whether a strategic rebalancing toward more profitable segments is warranted.

Customer retention analysis shows significant variation in purchase frequency across the customer base. Identifying high-frequency, high-value customers and developing targeted loyalty programs to strengthen these relationships would protect the revenue base. Simultaneously, analyzing the characteristics of churned customers would enable development of targeted re-engagement strategies.

6.7 Correlation Insights

- The weak positive correlation between discount and sales ($r = 0.03$) confirms that discounts are not effectively driving incremental demand — the primary motivation for discounting — while the strong negative correlation with profit ($r = -0.22$) confirms their profitability-destructive effect
- The moderate positive correlation between sales and profit ($r = 0.48$) indicates that revenue growth generally translates to profit growth, but with high variability driven by the discount factor
- Quantity and sales show a moderate positive correlation, confirming that bulk orders drive higher revenue but not necessarily proportionately higher profitability due to quantity-based discount expectations

6.8 Business Problems Identified

- Systematic over-discounting in the Furniture category destroying profitability
- Tables sub-category generating consistent losses across all markets — potential candidate for portfolio exit or fundamental repricing

- Central region chronic underperformance requiring strategic review
- Ineffective discount strategy: discounts are being applied broadly without evidence of generating meaningful incremental demand
- Revenue seasonality concentration in Q4 creating operational and cash flow management challenges
- Imbalance between Corporate account revenue contribution and profitability

6.9 Recommendations

- Implement a tiered discount authorization framework requiring management approval for all discounts exceeding 20%
- Conduct a comprehensive profitability review of all Furniture sub-categories, particularly Tables, with target of achieving minimum 5% margin within 12 months
- Launch a Central region performance improvement initiative with state-level analysis of competitive dynamics and cost drivers
- Develop Q4 demand forecasting capability to optimize inventory positioning and prevent stockouts of high-demand items during peak season
- Establish a key account profitability review process for the top 20 Corporate accounts to ensure adequate margin on negotiated contracts
- Implement product bundling strategies for high co-occurrence product combinations identified in the analysis
- Create a systematic process for identifying and addressing chronically loss-making products, including repricing, cost reduction, or portfolio exit decisions

CHAPTER 7: RESULTS & DISCUSSION

This chapter presents a critical evaluation of the results generated by the EDA process, examining their reliability, significance, and practical implications. The discussion contextualizes the analytical findings within the broader business environment, compares observed patterns with theoretical expectations, identifies the limitations of the analysis, and articulates the pathways through which the insights can be operationalized to drive business improvement.

7.1 Results of Data Analysis

The comprehensive EDA conducted on the Superstore dataset generated a rich set of analytical results across multiple business dimensions. From a sales performance perspective, the analysis confirmed healthy revenue growth over the four-year period with consistent seasonal patterns. From a profitability perspective, the analysis revealed significant heterogeneity in profit margins across product categories, geographic regions, customer segments, and discount tiers. From a behavioral perspective, the analysis uncovered distinct patterns in customer purchasing behavior across segments and geographies.

The statistical robustness of the results is supported by the large dataset size (9,994 transactions), which provides adequate statistical power for detecting meaningful differences across analytical segments. The consistency of patterns across multiple years of data further strengthens confidence in the findings, as persistent patterns are less likely to represent random variation than single-period observations.

7.2 Interpretation of Results

The central analytical finding — the systematic negative relationship between discount depth and profitability — has clear and direct business implications. The data unambiguously demonstrates that the current approach to discounting is not generating adequate sales uplift to justify its cost in margin compression. This finding is consistent with extensive academic research on promotional elasticity in retail, which generally finds that price promotions are most effective for stimulating

trial of new products or defending market share against competitive threats, but are ineffective as a sustained revenue growth strategy.

The regional performance disparities are more complex to interpret as they likely reflect a combination of factors including competitive intensity, customer income levels, shipping costs, and market development stage. Disentangling these factors would require additional data sources and more sophisticated analytical approaches, but the EDA results clearly identify the existence and magnitude of these disparities and provide the foundation for more targeted investigation.

7.3 Comparison with Expected Outcomes

The results of the analysis are broadly consistent with theoretical expectations and patterns commonly documented in the retail analytics literature. The negative discount-profit relationship aligns with canonical retail economics theory. The Pareto concentration of revenue in a small proportion of products is a well-documented retail phenomenon. The Q4 seasonal pattern is characteristic of US retail businesses. These consistencies with established patterns provide additional confidence in the analytical methodology and results.

One finding that warrants further investigation is the relatively weak positive correlation between discount and sales ($r = 0.03$). Standard microeconomic theory and promotional elasticity research would predict a stronger positive relationship between discount depth and sales volume. The weak observed relationship suggests either that the Superstore customer base is relatively price-inelastic, that discounts are not being effectively communicated to customers, or that discounts are concentrated in low-demand periods when demand stimulation is inherently limited.

7.4 Impact on Business Decision-Making

- Pricing Decisions: The discount analysis provides direct and quantified guidance for repricing loss-making products and reducing excessive discount depth across categories
- Portfolio Management: The sub-category profitability analysis identifies specific products for repricing, cost reduction, or portfolio exit consideration

- Geographic Strategy: The regional performance analysis enables differentiated resource allocation based on market potential and current performance
- Customer Strategy: The segment profitability analysis informs customer relationship management priorities and account-level pricing strategy
- Inventory Management: The seasonal pattern analysis enables improved demand forecasting and inventory positioning
- Marketing Investment: The category and regional performance data enables more precise allocation of marketing budgets to highest-ROI opportunities

7.5 Discussion

This project demonstrates that EDA, rigorously applied to a comprehensive retail dataset, can generate insights that are both analytically rigorous and immediately actionable from a business strategy perspective. The combination of statistical analysis and intuitive visualization enables insights to be communicated effectively to both technical and non-technical audiences, bridging the gap between data science and business management.

The Streamlit dashboard developed as part of this project represents a practical mechanism for operationalizing these insights, enabling ongoing data-driven monitoring of business performance rather than limiting the analysis to a one-time retrospective study. The interactive filtering capabilities of the dashboard allow business users to investigate specific hypotheses and questions that arise from the high-level insights, supporting a culture of data-driven inquiry throughout the organization.

7.6 Challenges Faced

- Data quality issues including inconsistent date formats and a small number of missing values required careful handling to avoid introducing analytical bias
- The interpretation of regional performance differences required domain knowledge about US retail markets to avoid drawing overly simplistic causal conclusions

- Balancing visualization complexity and interpretability required iterative refinement to ensure charts were both informative and accessible to non-technical users
- Determining appropriate discount tier boundaries required domain judgment and sensitivity analysis to ensure the categorization was analytically meaningful
- Managing Streamlit session state for the theme toggle and filter interactions required careful implementation to ensure consistent application behavior

CHAPTER 8 : APPLICATIONS

8.1 Business Applications

The Business Insight Dashboard developed in this project has broad applicability across diverse business domains and organizational contexts. The core analytical capabilities of sales performance monitoring, profitability analysis, customer segmentation, and geographic performance assessment are relevant to virtually any business that generates transactional sales data.

1. Retail Industry

In the retail sector, businesses manage complex product portfolios across multiple locations and customer segments. The dashboard provides retail managers with the analytical tools needed to monitor total sales and profit performance, identify best-selling and loss-making products, analyze customer purchasing behavior across segments, optimize pricing and discount strategies, and plan inventory allocation based on seasonal demand patterns. Retail store managers can use the regional analysis to benchmark their location's performance against company-wide averages and identify specific areas for improvement.

2. E-commerce Platforms

E-commerce businesses generate massive volumes of transactional data through their digital sales channels. The dashboard can be adapted to analyze online order trends over time, evaluate the effectiveness of promotional discount campaigns, identify top-performing and underperforming product categories, understand customer segment behavior and lifetime value, and optimize product recommendation algorithms based on co-purchase patterns. The temporal trend analysis is particularly valuable for e-commerce businesses in planning flash sales and managing inventory during demand spikes.

3. Sales and Marketing Management

Sales and marketing teams can leverage the dashboard for performance management, territory planning, and campaign evaluation. Sales managers can use the regional breakdown to monitor territory performance, identify underperforming regions that require sales intervention, and allocate marketing resources based on data-driven assessments of market opportunity. Marketing teams can use customer segment analysis to develop targeted campaigns for the most valuable customer groups and evaluate the effectiveness of promotional strategies through before-and-after profit margin comparisons.

4. Financial Analysis and Reporting

Finance teams can use the dashboard as a profitability management tool, monitoring profit margins at product, category, and regional levels. The discount impact analysis provides direct evidence for financial decision-making about pricing authority and discount approval processes. The temporal trend analysis enables variance analysis between actual and budgeted performance across time periods. The ability to download filtered data in CSV format facilitates integration with formal financial reporting workflows.

5. Supply Chain and Operations Management

Supply chain and operations teams can leverage the analytical insights to optimize inventory planning, align production or procurement schedules with seasonal demand patterns, and identify the highest-velocity product SKUs that require priority availability management. The fulfillment efficiency analysis (based on order-to-ship timing) provides operational performance benchmarks and highlights opportunities for logistics improvement.

8.2 Real-world Use Cases

Use Case 1: Category Profitability Review — Furniture Manufacturer

A furniture manufacturing company suspects that certain product lines are generating inadequate returns despite strong revenue. Using the dashboard's profit analysis view, the management team identifies that the Tables sub-category generates a negative profit margin of -8.6%, primarily

driven by excessive promotional discounts. The insight leads to a strategic decision to implement a minimum price policy for Tables products and restrict discount authorization to products above a minimum margin threshold. Six months after implementation, the Tables sub-category achieves a positive margin for the first time in three years.

Use Case 2: Regional Expansion Planning — National Retailer

A national retailer is planning to expand its physical presence and wants to identify the most promising geographic markets. Using the dashboard's regional breakdown, the planning team identifies states with high sales volumes but below-average profit margins as markets requiring strategic attention before additional investment. They identify California as a high-opportunity market with both strong sales and profit performance and recommend prioritizing new locations in high-performing California cities. The dashboard's city-level analysis provides granular data supporting specific location selection decisions.

Use Case 3: Promotional Effectiveness Evaluation — Marketing Team

A marketing team has been running aggressive discount promotions for two quarters and wants to evaluate their effectiveness. Using the dashboard's discount impact analysis, the team quantifies that promotions with discounts above 35% generate average losses rather than profits, and that the correlation between discount depth and incremental sales volume is essentially zero. This data-driven insight leads to a fundamental redesign of the promotional strategy, replacing deep discount promotions with value-added bundles that maintain margins while providing customer incentives.

Use Case 4: Investor Presentation — Startup Retail Business

A retail startup needs to prepare financial performance data for an investor presentation. Using the dashboard, the founding team creates a comprehensive analytical narrative covering four years of business growth, demonstrating a 14.8% CAGR in revenue, identifying the key product categories driving profitability, and outlining data-driven strategic initiatives to address identified profitability challenges. The interactive dashboard is embedded in the presentation to demonstrate the company's analytical sophistication and data-driven management approach.

CHAPTER 9 : ADVANTAGES & LIMITATIONS

9.1 Advantages

1. End-to-End Integrated Pipeline

The most significant advantage of the proposed system is its provision of a complete end-to-end analytical pipeline from raw data through cleaning, analysis, visualization, and insight generation in a single integrated application. Users are not required to switch between multiple tools or manually transfer data between stages, eliminating inefficiency and the potential for data inconsistencies arising from tool switching.

2. Accessibility for Non-Technical Users

The Streamlit web interface provides access to sophisticated analytical capabilities without requiring any programming knowledge. Business users can explore the dashboard through intuitive point-and-click interactions, apply filters, and interpret visualizations without understanding the underlying Python code. This democratization of analytical capability significantly expands the potential user base beyond data specialists to include business managers, marketing professionals, and executives.

3. Interactive and Dynamic Analysis

All dashboard visualizations are fully interactive, supporting hover tooltips that reveal precise data values, zoom and pan capabilities for detailed examination of specific data regions, and dynamic updating when filters are changed. This interactivity enables users to investigate specific hypotheses and questions that arise during data exploration, supporting a genuine analytical dialogue with the data rather than static report consumption.

4. Flexible Data Compatibility

The schema validation and column mapping system enables the dashboard to work with a wide variety of retail datasets beyond the reference Superstore dataset. The generic explorer mode

provides a fallback analytical experience for completely non-standard datasets, ensuring that the application provides value even when input data does not conform to the expected schema.

5. Dual Theme Support

The light and dark theme options accommodate different user preferences and working environments. The dark theme is particularly valuable for extended analysis sessions in low-light environments, reducing eye strain during prolonged data exploration. The theme preference is preserved in session state, maintaining consistency throughout the analytical session.

6. Open Source and Customizable

The complete implementation in Python using open-source libraries means that the application can be freely used, modified, and extended without licensing costs. Organizations can customize the dashboard to incorporate their specific business metrics, branding, and analytical requirements. The clean, well-commented code structure facilitates understanding and modification by developers.

9.2 Limitations

1. Dependency on Data Quality

The accuracy and reliability of the insights generated by the dashboard are fundamentally dependent on the quality of the input dataset. Systematic errors, biases, or gaps in the source data will propagate through the analysis and potentially generate misleading insights. While the data cleaning pipeline addresses common data quality issues, it cannot detect or correct all possible data quality problems, particularly those arising from incorrect data entry or systematic measurement errors in the source systems.

2. CSV Format Limitation

The current implementation accepts data exclusively in CSV format. Organizations that store their operational data in relational databases, cloud data warehouses, or other file formats (Excel, JSON, Parquet) cannot connect the dashboard directly to their data sources without first exporting to CSV.

This limitation creates an additional step in the workflow and introduces the risk of working with stale data exports rather than real-time operational data.

3. Performance with Large Datasets

The Pandas-based data processing pipeline is implemented in-memory, meaning that all data must fit within the available RAM of the server running the application. While the current implementation handles datasets of tens of thousands to low hundreds of thousands of rows efficiently, performance degrades significantly with very large datasets (millions of rows). Organizations with high-volume transactional data would need to implement data sampling or aggregation strategies to maintain acceptable performance.

4. Absence of Predictive Analytics

The dashboard is exclusively focused on descriptive and exploratory analytics — understanding what has happened and why. It does not include predictive modeling capabilities such as sales forecasting, demand prediction, or customer churn propensity scoring. These predictive capabilities represent a significant analytical opportunity that would substantially enhance the business value of the platform.

5. Single User Session Architecture

The Streamlit application architecture does not include multi-user authentication or role-based access control. All users who access the application see the same data and have the same permissions. Organizations requiring differentiated access control (e.g., regional managers seeing only their region's data) would need to implement additional authentication and data filtering logic.

CHAPTER 10 : FUTURE SCOPE

10.1 Future Enhancements

1. Database and Cloud Data Integration

A priority future enhancement is the addition of native connectivity to relational databases (MySQL, PostgreSQL, SQL Server), cloud data warehouses (Google BigQuery, Amazon Redshift, Snowflake), and cloud storage services (AWS S3, Google Cloud Storage, Azure Blob Storage). This would eliminate the CSV export requirement and enable the dashboard to work directly with live operational data. Database connectivity would also enable the analysis of much larger datasets through query-based data extraction and aggregation.

2. Machine Learning and Predictive Analytics

The addition of machine learning capabilities would substantially expand the analytical value of the platform. Priority predictive features include: sales forecasting using time series models (ARIMA, Prophet, or LSTM neural networks) to predict future sales volumes and seasonal patterns; customer lifetime value prediction to identify high-potential customers for targeted retention investment; churn prediction to identify customers at risk of defection; demand forecasting for inventory optimization; and anomaly detection to identify unusual sales patterns that may indicate fraud or system errors.

3. Multi-User Authentication and Role-Based Access

Implementing a user authentication system with role-based access control would enable secure multi-user deployment in organizational settings. Roles could include Executive (full data access), Regional Manager (data filtered to their region), Category Manager (data filtered to their product categories), and Analyst (full data access with export capabilities). Integration with enterprise authentication systems (Active Directory, OAuth2, SAML) would enable seamless deployment within existing organizational IT infrastructure.

Real-Time Data Streaming

4. Integration with real-time data streaming platforms such as Apache Kafka or cloud-native streaming services would enable the dashboard to display live business performance metrics that update in real time as transactions are processed. This would transform the dashboard from a retrospective analytical tool into an operational monitoring platform capable of detecting and alerting on unusual business conditions as they occur.

5. Natural Language Query Interface

Implementing a natural language query interface using large language model (LLM) technology would enable business users to ask analytical questions in plain English and receive automatically generated visualizations and insights. For example, a user could type 'Show me the profit margin trend for Technology products in the West region over the last two years' and receive an automatically generated, appropriate visualization. This NLQ capability would further lower the analytical barrier for non-technical business users.

6. Automated Report Generation

Adding automated report generation capabilities would enable the dashboard to create professionally formatted PDF or PowerPoint reports summarizing key analytical findings, trends, and recommendations. These reports could be scheduled for automatic generation and email distribution to relevant stakeholders on a daily, weekly, or monthly basis, ensuring consistent data-driven communication throughout the organization without requiring manual report creation effort.

7. Mobile Responsive Design

Redesigning the user interface for full mobile responsiveness would enable business users to access analytical insights on smartphones and tablets. Given the increasing prevalence of mobile work patterns, particularly among field sales and regional management roles, mobile accessibility represents an important usability enhancement. The redesign would adapt the layout, chart sizes, and interaction patterns to work effectively on small touchscreen displays.

8. Advanced Visualization Types

Additional visualization types would enrich the analytical capabilities of the platform. Geographic map visualizations using choropleth maps would provide intuitive representation of regional and state-level performance variations. Sankey diagrams would visualize customer journey and product flow patterns. Network graphs would visualize product co-purchase relationships. Waterfall charts would effectively communicate profit bridge analysis. Animated visualizations would enable temporal evolution of business metrics to be displayed dynamically.

9. A/B Testing and Experiment Analysis

Adding capabilities for analyzing the results of pricing experiments, promotional campaigns, and other business interventions using A/B testing statistical frameworks would enable the dashboard to directly support evidence-based business experimentation. This would include automatic computation of statistical significance, confidence intervals, and minimum detectable effect sizes for business metric comparisons between experimental and control conditions.

10. Multi-Language and Internationalization Support

Adding support for multiple display languages and international number and date formatting conventions would enable deployment of the dashboard in global organizational contexts. This would include translation of all UI text elements, support for locale-specific number formats (comma vs. period as decimal separator), date format variations, and currency symbol display.

CHAPTER 11: CONCLUSION

11.1 Summary

This project has successfully achieved all of its stated objectives in developing a comprehensive, systematic, and practically valuable framework for generating business insights from retail sales data using Exploratory Data Analysis. The project has demonstrated the complete EDA lifecycle from raw data ingestion through cleaning, preprocessing, multi-dimensional statistical analysis, visualization, and structured insight generation, culminating in the delivery of an interactive web-based business intelligence dashboard.

The analytical work conducted on the Superstore retail dataset has generated actionable insights across five critical business dimensions: sales performance and growth, product and category profitability, pricing and discount strategy effectiveness, geographic market performance, and customer segment behavior. These insights are grounded in rigorous statistical analysis of nearly 10,000 transaction records spanning four years of business operations, providing a robust empirical foundation for strategic decision-making.

The technical implementation demonstrates proficiency in the modern Python data science ecosystem, effectively leveraging Pandas for data manipulation, NumPy for numerical computation, Plotly for interactive visualization, and Streamlit for web application development. The resulting dashboard represents a production-quality analytical tool that could be deployed and used effectively in a real business environment, not merely an academic demonstration project.

The literature review confirmed that EDA remains the foundational methodology for data-driven business analysis despite the emergence of sophisticated machine learning and artificial intelligence techniques. The ability to systematically understand historical data patterns, detect anomalies, identify relationships between variables, and generate human-interpretable insights from that understanding remains the essential prerequisite for all higher-order analytical activities.

11.2 Final Outcome

The Business Insight Dashboard delivered as the primary outcome of this project provides a genuinely useful analytical tool for retail business management. The dashboard enables business users to monitor key performance metrics in real time, drill into specific dimensions of business performance, investigate the drivers of profitability and loss, and export filtered data for further analysis. The interactive nature of the dashboard supports a genuine analytical dialogue between the user and the data, enabling progressive hypothesis investigation and insight discovery.

From an academic perspective, this project has provided the team with comprehensive practical experience across the full data science workflow, from data engineering through statistical analysis to application development and deployment. The skills developed in this project — data manipulation with Pandas, statistical analysis, data visualization with Plotly, web application development with Streamlit, and systematic analytical thinking — represent a highly transferable and commercially valuable competency set that will serve the team throughout their professional careers.

Looking forward, the project has established a solid foundation for the future enhancements outlined in Chapter 14. The clean, modular code architecture facilitates the addition of new features and capabilities, and the open-source implementation enables community contributions. The integration of machine learning capabilities, real-time data connectivity, and natural language query interfaces represents a compelling roadmap for evolving the current analytical dashboard into a comprehensive business intelligence platform capable of serving the analytical needs of complex, large-scale retail organizations.

In conclusion, this project affirms that Exploratory Data Analysis, rigorously applied and thoughtfully communicated, remains one of the most powerful and accessible tools available for generating business intelligence. The combination of statistical rigor, intuitive visualization, and structured insight narration embodied in this project represents a model for effective data-driven business analysis that is both academically sound and practically valuable.

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APPENDIX

Complete Source Code

```
import pandas as pd
import plotly.express as px
import plotly.graph_objects as go
import streamlit as st
from pathlib import Path

st.set_page_config(page_title="Business Insight Dashboard", layout="wide")

if "theme" not in st.session_state:
    st.session_state.theme = "light"

st.markdown(
    '<div style="background: linear-gradient(135deg, #7c3aed 0%, #ec4899 100%);text-align: center;padding: 3rem 2rem;border-radius: 24px;margin-bottom: 2rem;box-shadow: 0 20px 50px rgba(124, 58, 237, 0.25);">'
    '<h1 style="margin: 0; color: white; font-size: 2.8rem; font-weight: 800; letter-spacing: -1px;">Insights Dashboard</h1>'
    '<p style="margin: 0.8rem 0 0 0; color: rgba(255,255,255,0.95); font-size: 1.1rem; font-weight: 400;">Visualize patterns. Make better decisions.</p>'
    "</div>",
    unsafe_allow_html=True,
)

THEMES = {
    "light": {
        "primary": "#7c3aed",
        "secondary": "#ec4899",
        "bg_main": "#ffffff",
        "bg_secondary": "#f9fafb",
        "text_primary": "#111827",
        "text_secondary": "#6b7280",
        "border": "#e5e7eb",
        "card_shadow": "0 4px 20px rgba(0,0,0,0.08)",
        "expander_bg": "#f8fafc",
    },
    "dark": {
        "primary": "#a78bfa",
        "secondary": "#f472b6",
        "bg_main": "#1f2937",
        "bg_secondary": "#111827",
        "text_primary": "#f9fafb",
        "text_secondary": "#d1d5db",
```

```
        "border": "#374151",
        "card_shadow": "0 4px 20px rgba(0,0,0,0.3)",
        "expander_bg": "#111827",
    },
}

BASE_CSS = """
<style>
@import
url('https://fonts.googleapis.com/css2?family=Poppins:wght@400;500;600;700;800&display=sw
ap');
:root{{--primary:{primary};--secondary:{secondary};--bg-main:{bg_main};--bg-
secondary:{bg_secondary};--text-primary:{text_primary};--text-secondary:{text_secondary};--
border:{border};--card-shadow:{card_shadow};--expander-bg:{expander_bg};}}
*{{font-family:'Poppins',-apple-system,sans-serif;}}
body,main{{background:var(--bg-secondary)!important;}}
.block-container{{padding:2rem 3rem!important;}}
[data-testid="stMetric"]{{background:linear-gradient(135deg,#7c3aed 0%,#ec4899
100%)!important;padding:16px 18px!important;border-
radius:16px!important;color:white!important;box-shadow:var(--card-
shadow)!important;border:none!important;}}
[data-testid="stMetric"] label{{color:rgba(255,255,255,0.95)!important;font-
size:0.7rem!important;font-weight:600!important;line-height:1.1!important;white-
space:normal!important;}}
[data-testid="stMetric"] [data-testid="stMetricValue"]{{color:white!important;font-
size:1.4rem!important;font-weight:800!important;line-height:1.15!important;white-
space:normal!important;}}
div[data-testid="stSidebar"]{{background:var(--bg-main)!important;border-right:1px solid var(--
border)!important;}}
div[data-testid="stSidebar"] *,div[data-testid="stSidebar"] [data-testid="stMarkdownContainer"]
*{{color:var(--text-primary)!important;opacity:1!important;}}
h1,h2,h3,h4,h5,h6,div[data-testid="stMarkdownContainer"],div[data-
testid="stMarkdownContainer"] h1,div[data-testid="stMarkdownContainer"] h2,div[data-
testid="stMarkdownContainer"] h3,div[data-testid="stMarkdownContainer"] h4,div[data-
testid="stMarkdownContainer"] h5,div[data-testid="stMarkdownContainer"] h6,div[data-
testid="stMarkdownContainer"] p,div[data-testid="stMarkdownContainer"] li,div[data-
testid="stMarkdownContainer"] span{{color:var(--text-
primary)!important;opacity:1!important;}}
h3{{font-size:1.3rem!important;margin-top:1.5rem!important;}}
[data-testid="stSelectbox"],[data-testid="stMultiSelect"],[data-testid="stRadio"],[data-
testid="stSlider"],[data-testid="stTextArea"]{{padding:12px 14px!important;border-
radius:12px!important;border:2px solid var(--border)!important;background:var(--bg-
main)!important;color:var(--text-primary)!important;}}
.stButton>button{{background:linear-gradient(135deg,#7c3aed 0%,#ec4899
100%)!important;color:white!important;border:none!important;border-
```

```
radius:10px!important;padding:12px 28px!important;font-weight:600!important;box-shadow:0 4px 15px rgba(124,58,237,0.25)!important;}}
.stExpander,div[data-testid="stSidebar"] .stExpander{{border:2px solid var(--border)!important;border-radius:12px!important;background:var(--expander-bg)!important;}}
div[data-testid="stSidebar"] .stExpander summary,div[data-testid="stSidebar"] .stExpander summary *,div[data-testid="stSidebar"] .stExpander svg{{color:var(--text-primary)!important;opacity:1!important;}}
.stDataFrame{{border:1px solid var(--border)!important;border-radius:12px!important;background:var(--bg-main)!important;}}
.stDataFrame [role="gridcell"],.stDataFrame [role="columnheader"]{{font-size:0.78rem!important;line-height:1.2!important;}}
.stDataFrame [role="gridcell"]{{white-space:normal!important;word-break:break-word!important;}}
</style>
"""

theme = THEMES.get(st.session_state.theme, THEMES["light"])
st.markdown(BASE_CSS.format(**theme), unsafe_allow_html=True)

USE_COLUMN_CONFIG = hasattr(st, "column_config")

def show_fig(fig, y_reversed=False, **layout):
    fig.update_layout(template="plotly_white", **layout)
    if y_reversed:
        fig.update_yaxes(autorange="reversed")
    st.plotly_chart(fig, use_container_width=True)

TEXT_COLS = {"Order ID": "medium", "Ship Mode": "small", "Customer ID": "medium", "Customer Name": "medium", "Segment": "small", "Country": "small", "City": "medium", "State": "medium", "Postal Code": "small", "Region": "small", "Product ID": "medium", "Category": "small", "Sub-Category": "medium", "Product Name": "large"}
DATE_COLS = {"Order Date": "YYYY-MM-DD", "Ship Date": "YYYY-MM-DD"}
NUM_COLS = {"Sales": "$,.2f", "Quantity": ",", "Discount": ".0%", "Profit": "$,.2f"}

def render_filtered_table(data):
    display_df = data.copy()
    for col in ["Order Date", "Ship Date"]:
        if col in display_df.columns:
            display_df[col] = pd.to_datetime(display_df[col], errors="coerce").dt.date
    if "Postal Code" in display_df.columns:
        display_df["Postal Code"] = display_df["Postal Code"].fillna("").astype(str).str.replace(r"\.0$", "", regex=True)
    st.markdown("---")
    st.subheader("Filtered Data")
    st.caption("Scroll to view all columns. Use the search box in the table header to find rows quickly.")
```

```
    table_height = min(760, max(360, 40 + len(display_df) * 26))
    if USE_COLUMN_CONFIG:
        cols = display_df.columns
        column_config = {k: st.column_config.TextColumn(k, width=v) for k, v in
TEXT_COLS.items() if k in cols}
        column_config.update({k: st.column_config.DateColumn(k, format=v, width="small") for k,
v in DATE_COLS.items() if k in cols})
        column_config.update({k: st.column_config.NumberColumn(k, format=v) for k, v in
NUM_COLS.items() if k in cols})
        st.dataframe(display_df, use_container_width=True, height=table_height, hide_index=True,
column_config=column_config)
    else:
        format_dict = {k: v for k, v in NUM_COLS.items() if k in display_df.columns}
        st.dataframe(display_df.style.format(format_dict), use_container_width=True,
height=table_height)
    csv = display_df.to_csv(index=False).encode("utf-8")
    st.download_button("Download filtered data", csv, file_name="filtered_data.csv",
mime="text/csv")

@st.cache_data
def load_data(uploaded_file=None):
    if uploaded_file is None:
        df = pd.read_csv("Sample - Superstore.csv", encoding="latin1")
    else:
        try:
            df = pd.read_csv(uploaded_file, encoding="latin1")
        except Exception:
            st.error("Could not read the uploaded file. Please make sure it's a valid CSV with the same
structure as the sample.")
            return pd.DataFrame()
    for col in ["Order Date", "Ship Date"]:
        if col in df.columns:
            df[col] = pd.to_datetime(df[col], errors="coerce")
    if "Order Date" in df.columns:
        df["Year"] = df["Order Date"].dt.year
        df["Month"] = df["Order Date"].dt.to_period("M").astype(str)
    return df

df = load_data(st.sidebar.file_uploader("Upload CSV (optional)", type=["csv"]))

if df.empty:
    st.error("No data available. Please upload a valid CSV or check the sample file.")
    st.stop()

required = ["Sales", "Profit", "Order ID"]
generic_mode = any(c not in df.columns for c in required)
```

```
with st.sidebar:
    st.markdown("##  Quick Guide")
    with st.expander("How to use this dashboard", expanded=True):
        st.markdown("1. **Upload your own file** or use the built-in sample dataset.\n2. Choose filters below; you can select multiple values for each attribute.\n3. Pick a view and explore interactive charts. Hover and click for details.\n4. Beginners: start with the **Overview** tab before diving deeper.\n5. If your file doesn't fit the expected schema, the app will switch to a generic explorer mode with basic summaries and histograms.")
    st.markdown("---")
    st.markdown("#### Theme")
    col1, col2 = st.columns(2)
    if col1.button(" Light", use_container_width=True):
        st.session_state.theme = "light"
        st.rerun()
    if col2.button("Dark", use_container_width=True):
        st.session_state.theme = "dark"
        st.rerun()
    st.markdown("---")
    if generic_mode:
        st.markdown("##  Generic filters")
        filter_col = st.selectbox("Pick a column to filter (optional)", options=[""] + df.columns.tolist())
        selected_values = st.multiselect(f"Values for {filter_col}", options=sorted(df[filter_col].dropna().unique().tolist())) if filter_col else []
        generic_filter = (filter_col, selected_values)
        view_option = st.radio(" Choose a view", options=["Overview"])
    else:
        st.markdown("##  Filters")

    def options_for(col, warn_key):
        if col not in df.columns:
            st.warning(f"Dataset does not contain a '{col}' column; {warn_key} filtering will be skipped.")
            return []
        return sorted(df[col].dropna().unique().tolist())

    filter_defs = [
        ("Year", " Year (pick one or more, blank = all)", "year", lambda opts: opts if len(opts) <= 3 else []),
        ("Region", " Region (pick one or more)", "region", None),
        ("Category", " Category (pick one or more)", "category", None),
        ("Segment", "Segment (pick one or more)", "segment", None),
    ]
    selections = {}
    for col, label, warn_key, default_fn in filter_defs:
```

```
        opts = options_for(col, warn_key)
        selections[col] = st.multiselect(label, options=opts, default=(default_fn(opts) if default_fn else []))
    selected_year, selected_region = selections["Year"], selections["Region"]
    selected_category, selected_segment = selections["Category"], selections["Segment"]
    st.markdown("---")

    view_option = st.radio(
        " Choose a view",
        options=["Overview", "Top Products", "Sales Trends", "Profit Analysis", "Regional Breakdown"],
    )

    with st.expander(" Feedback", expanded=False):
        with st.form("feedback_form", clear_on_submit=True):
            rating = st.slider("Rate this dashboard", 1, 5, 3)
            comments = st.text_area("Comments (optional)")
            submitted = st.form_submit_button("Submit")
            if submitted:
                try:
                    import datetime
                    fb = pd.DataFrame([{"timestamp": datetime.datetime.now(), "rating": rating, "comments": comments}])
                    fb.to_csv("feedback.csv", mode="a", header=not pd.io.common.file_exists("feedback.csv"), index=False)
                    st.success("Thanks for your feedback!")
                except Exception:
                    st.error("Could not save feedback.")

        admin_pin = st.secrets.get("ADMIN_PIN", "")
        show_admin = st.toggle("Admin View", value=False)
        if show_admin:
            if not admin_pin:
                st.info("Admin view is disabled. Add ADMIN_PIN in Streamlit secrets.")
            else:
                pin_input = st.text_input("Admin PIN (private)", type="password", key="admin_pin_input")
                if pin_input:
                    if pin_input == admin_pin:
                        feedback_path = Path("feedback.csv")
                        if feedback_path.exists():
                            try:
                                fb_df = pd.read_csv(feedback_path)
                                if "timestamp" in fb_df.columns:
                                    fb_df["timestamp"] = pd.to_datetime(fb_df["timestamp"], errors="coerce")
                                    fb_df = fb_df.sort_values("timestamp", ascending=False)
```

```
                st.markdown("##### Recent Feedback")
                st.dataframe(fb_df.head(50), use_container_width=True, height=220)
            except Exception:
                st.info("Feedback saved. Unable to load the table.")
        else:
            st.caption("No feedback yet.")
    else:
        st.error("Invalid PIN.")
else:
    st.caption("Enter the admin PIN to view feedback.")

filtered = df.copy()
if generic_mode:
    col, vals = generic_filter
    if col and vals:
        filtered = filtered[filtered[col].isin(vals)]
else:
    for col_name, selected in [("Year", selected_year), ("Region", selected_region), ("Category",
selected_category), ("Segment", selected_segment)]:
        if selected and col_name in filtered.columns:
            filtered = filtered[filtered[col_name].isin(selected)]

if filtered.empty:
    st.warning("No records match the selected filters. Please adjust the
year/region/category/segment selections or upload a different dataset.")

if generic_mode:
    st.markdown("### Simple Dataset Summary")
    num_cols = filtered.select_dtypes(include=["number"]).columns.tolist()
    cat_cols = filtered.select_dtypes(exclude=["number"]).columns.tolist()

    def first_col(columns, keywords):
        return next((c for c in columns if any(k in c.lower() for k in keywords)), None)

    def format_number(val):
        if pd.isna(val):
            return "—"
        return f"{val:,.0f}" if abs(val) >= 1000 else f"{val:,.2f}"

    brand_col = first_col(cat_cols, ["brand", "company", "manufacturer", "make"])
    if not brand_col and cat_cols:
        brand_col = next((c for c in cat_cols if filtered[c].nunique() <= 20), cat_cols[0])

    top_brands = []
    if brand_col:
        try:
```

```
        top_brands = filtered[brand_col].dropna().astype(str).value_counts().index.tolist()[:3]
    except Exception:
        pass

ram_col = first_col(num_cols, ["ram", "memory"])
best_ram = "—"
if ram_col:
    max_ram = filtered[ram_col].max(skipna=True)
    if not pd.isna(max_ram):
        col_lower = ram_col.lower()
        if "gb" in col_lower:
            best_ram = f"{format_number(max_ram)} GB"
        elif "mb" in col_lower:
            best_ram = f"{format_number(max_ram)} MB"
        elif max_ram <= 256:
            best_ram = f"{format_number(max_ram)} GB"
        else:
            best_ram = format_number(max_ram)

price_col = first_col(num_cols, ["price", "cost", "amount", "revenue", "sales"])
avg_price = "—"
if price_col:
    avg_val = filtered[price_col].mean(skipna=True)
    if not pd.isna(avg_val):
        avg_price = f"${avg_val:,.0f}" if any(k in price_col.lower() for k in ["price", "cost"]) else
format_number(avg_val)

labels = ["Most Selling Brand", "2nd Most Selling", "3rd Most Selling"]
metric_items = [(labels[i], top_brands[i]) for i in range(min(3, len(top_brands)))]
if best_ram not in ["", "—", None]:
    metric_items.append(("Best RAM", best_ram))
if avg_price not in ["", "—", None]:
    metric_items.append(("Average Price", avg_price))

for col, (label, value) in zip(st.columns(4), metric_items[:4]):
    col.metric(label, value)

st.markdown("---")
with st.expander("View sample rows", expanded=False):
    st.dataframe(filtered.head(8), use_container_width=True)
st.markdown("---")

palette = ["#FF6B6B", "#4D96FF", "#6BCB77", "#FFD93D", "#9D4EDD", "#FF8FAB",
"#06D6A0", "#F72585"]
if num_cols:
    st.subheader("Numeric distributions")
```

```
        sorted_num = sorted(num_cols, key=lambda c: filtered[c].count(), reverse=True)[:3]
        for i, col in enumerate(sorted_num):
            fig = px.histogram(filtered, x=col, nbins=30, title=f"Distribution of {col}",
color_discrete_sequence=[palette[i % len(palette)]])
            show_fig(fig)
    else:
        st.info("No numeric columns found to show distributions.")

    if cat_cols:
        st.subheader("Top categories (by frequency)")
        cand = [c for c in cat_cols if filtered[c].nunique() <= 20][:3]
        for col in cand:
            df_top = filtered[col].fillna("(missing)").value_counts().reset_index()
            df_top.columns = [col, "count"]
            fig_bar = px.bar(df_top.head(10), x="count", y=col, orientation="h", title=f"Top values in
{col}", color_discrete_sequence=palette)
            show_fig(fig_bar, y_reversed=True)
            fig_pie = px.pie(df_top.head(6), values="count", names=col, title=f"{col} composition
(top 6)", color_discrete_sequence=palette)
            show_fig(fig_pie)
    else:
        st.info("No categorical columns found to show category breakdowns.")
    st.stop()

# ======================== OVERVIEW ========================
if view_option == "Overview":
    st.markdown("### Overview")
    if "Sales" not in filtered.columns:
        st.error("Cannot display overview: 'Sales' column missing.")
    else:
        total_sales = filtered["Sales"].sum()
        total_profit = filtered["Profit"].sum() if "Profit" in filtered.columns else 0
        total_orders = filtered["Order ID"].nunique() if "Order ID" in filtered.columns else
len(filtered)
        avg_discount = filtered["Discount"].mean() * 100 if "Discount" in filtered.columns else None
        profit_margin = (total_profit / total_sales * 100) if total_sales else 0
    metrics = [
        ("Total Sales", f"${total_sales:,.0f}"),
        ("Total Profit", f"${total_profit:,.0f}"),
        ("Orders", f"{total_orders:,}"),
        ("Profit Margin", f"{profit_margin:.1f}%"),
        ("Avg Discount", "N/A" if avg_discount is None else f"{avg_discount:.1f}%"),
    ]
    for col, (label, value) in zip(st.columns(5), metrics):
        col.metric(label, value)
    st.markdown("####")
```

```
    col1, col2 = st.columns(2)
    with col1:
        if "Category" in filtered.columns:
            cat_sales = filtered.groupby("Category")[["Sales", "Profit"]].sum().reset_index()
            fig = px.bar(cat_sales, x="Category", y=["Sales", "Profit"], barmode="group", title="Sales
& Profit by Category", color_discrete_sequence=["#667eea", "#764ba2"])
            show_fig(fig, height=400, legend_title_text="")
        else:
            st.info("Category column not available; skipping category breakdown.")
    with col2:
        if "Segment" in filtered.columns:
            seg_sales = filtered.groupby("Segment")["Sales"].sum().reset_index()
            fig = px.pie(seg_sales, values="Sales", names="Segment", title="Sales Distribution by
Segment", color_discrete_sequence=["#667eea", "#764ba2", "#f093fb"], hole=0.4)
            show_fig(fig, height=400)
        else:
            st.info("Segment column not available; skipping segment distribution.")
    col3, col4 = st.columns(2)
    with col3:
        if "Region" in filtered.columns:
            region_data = filtered.groupby("Region")[["Sales", "Profit"]].sum().reset_index()
            fig = px.bar(region_data, x="Region", y=["Sales", "Profit"], barmode="group",
title="Sales & Profit by Region", color_discrete_sequence=["#667eea", "#764ba2"])
            show_fig(fig, height=400)
        else:
            st.info("Region column not available; skipping regional breakdown.")
    with col4:
        if "Ship Mode" in filtered.columns:
            ship_data = filtered.groupby("Ship Mode")["Sales"].sum().reset_index()
            fig = px.pie(ship_data, values="Sales", names="Ship Mode", title="Sales by Ship Mode",
color_discrete_sequence=["#667eea", "#764ba2", "#f093fb", "#a8edea"], hole=0.4)
            show_fig(fig, height=400)
        else:
            st.info("Ship Mode column not available; skipping shipping analysis.")


# ======================== TOP PRODUCTS ========================
elif view_option == "Top Products":
    st.markdown("### Top Products")
    if "Sales" not in filtered.columns:
        st.error("Cannot show top products: 'Sales' column missing.")
    else:
        n = st.slider("Number of products to display", 5, 20, 10)
    col1, col2 = st.columns(2)
    with col1:
        top_sales = filtered.groupby("Product Name")["Sales"].sum().nlargest(n).reset_index()
```

```
        fig = px.bar(top_sales, x="Sales", y="Product Name", orientation="h", title=f"Top {n} Products by Sales", color="Sales", color_continuous_scale="Purples")
        show_fig(fig, height=500, y_reversed=True, showlegend=False)
    with col2:
        top_profit = filtered.groupby("Product Name")["Profit"].sum().nlargest(n).reset_index()
        fig = px.bar(top_profit, x="Profit", y="Product Name", orientation="h", title=f"Top {n} Products by Profit", color="Profit", color_continuous_scale="Greens")
        show_fig(fig, height=500, y_reversed=True, showlegend=False)
    st.markdown("---")
    st.subheader("Sub-Category Performance")
    subcat = filtered.groupby("Sub-Category")[["Sales", "Profit"]].sum().reset_index().sort_values("Sales", ascending=False)
    fig = px.bar(subcat, x="Sub-Category", y=["Sales", "Profit"], barmode="group", title="Sales & Profit by Sub-Category", color_discrete_sequence=["#667eea", "#764ba2"])
    show_fig(fig, height=450, xaxis_tickangle=-45)


# ======================= SALES TRENDS =======================
elif view_option == "Sales Trends":
    st.markdown("### Sales Trends")
    if "Order Date" not in filtered.columns or "Sales" not in filtered.columns:
        st.error("Cannot show sales trends: 'Order Date' or 'Sales' column missing.")
    else:
        monthly = filtered.set_index("Order Date").resample("M")[["Sales", "Profit"]].sum().reset_index()
    fig = go.Figure()
    fig.add_trace(go.Scatter(
        x=monthly["Order Date"], y=monthly["Sales"],
        mode="lines+markers", name="Sales",
        line=dict(color="#667eea", width=2.5),
        marker=dict(size=5),
    ))
    fig.add_trace(go.Scatter(
        x=monthly["Order Date"], y=monthly["Profit"],
        mode="lines+markers", name="Profit",
        line=dict(color="#764ba2", width=2.5),
        marker=dict(size=5),
    ))
    show_fig(fig, height=450, title="Monthly Sales & Profit Trend", xaxis_title="Date", yaxis_title="Amount ($)")
    col1, col2 = st.columns(2)
    with col1:
        yearly = filtered.set_index("Order Date").resample("Y")[["Sales"]].sum().reset_index()
        yearly["Year"] = yearly["Order Date"].dt.year
        fig = px.bar(yearly, x="Year", y="Sales", title="Yearly Sales", color="Sales", color_continuous_scale="Purples", text_auto=",.0f")
```

```
        show_fig(fig, height=400, showlegend=False)
    with col2:
        yearly_p = filtered.set_index("Order Date").resample("Y")[["Profit"]].sum().reset_index()
        yearly_p["Year"] = yearly_p["Order Date"].dt.year
        fig = px.bar(yearly_p, x="Year", y="Profit", title="Yearly Profit", color="Profit",
color_continuous_scale="Greens", text_auto=",.0f")
        show_fig(fig, height=400, showlegend=False)
    st.markdown("---")
    st.subheader("Quarterly Breakdown")
    quarterly = filtered.set_index("Order Date").resample("Q")[["Sales",
"Profit"]].sum().reset_index()
    quarterly["Quarter"] = quarterly["Order Date"].dt.to_period("Q").astype(str)
    fig = px.bar(quarterly, x="Quarter", y=["Sales", "Profit"], barmode="group", title="Quarterly
Sales & Profit", color_discrete_sequence=["#667eea", "#764ba2"])
    show_fig(fig, height=400, xaxis_tickangle=-45)


# ======================== PROFIT ANALYSIS ========================
elif view_option == "Profit Analysis":
    st.markdown("### Profit Analysis")
    if "Profit" not in filtered.columns:
        st.error("Cannot show profit analysis: 'Profit' column missing.")
    else:
        col1, col2 = st.columns(2)
    with col1:
        fig = px.scatter(filtered, x="Sales", y="Profit", color="Category", title="Sales vs Profit",
color_discrete_sequence=["#667eea", "#764ba2", "#f093fb"], opacity=0.6, hover_data=["Product
Name"])
        show_fig(fig, height=450)
    with col2:
        fig = px.scatter(filtered, x="Discount", y="Profit", color="Category", title="Discount vs
Profit", color_discrete_sequence=["#667eea", "#764ba2", "#f093fb"], opacity=0.6)
        show_fig(fig, height=450)
    st.markdown("---")
    loss_products = filtered.groupby("Product
Name")["Profit"].sum().sort_values().head(10).reset_index()
    fig = px.bar(loss_products, x="Profit", y="Product Name", orientation="h", title="Top 10 Loss-
Making Products", color="Profit", color_continuous_scale="Reds_r")
    show_fig(fig, height=450, y_reversed=True, showlegend=False)
    st.markdown("---")
    st.subheader("Profit Margin by Sub-Category")
    subcat_pm = filtered.groupby("Sub-Category")[["Sales", "Profit"]].sum().reset_index()
    subcat_pm["Profit Margin %"] = (subcat_pm["Profit"] / subcat_pm["Sales"] * 100).round(1)
    subcat_pm = subcat_pm.sort_values("Profit Margin %", ascending=True)
    colors = ["#e74c3c" if v < 0 else "#667eea" for v in subcat_pm["Profit Margin %"]]
```

```
    fig = go.Figure(go.Bar(x=subcat_pm["Profit Margin %"], y=subcat_pm["Sub-Category"], orientation="h", marker_color=colors, text=subcat_pm["Profit Margin %"].apply(lambda x: f"{x:.1f}%"), textposition="outside"))
    show_fig(fig, height=450, title="Profit Margin % by Sub-Category", xaxis_title="Profit Margin %")


# ======================= REGIONAL BREAKDOWN =======================
elif view_option == "Regional Breakdown":
    st.markdown("### Geography view\nCompare performance across regions, states, and cities. The data table at the bottom gives an exact summary.")
    if "Region" not in filtered.columns or "Sales" not in filtered.columns:
        st.error("Cannot show regional breakdown: required columns missing.")
    else:
        region_summary = filtered.groupby("Region").agg(Sales=("Sales", "sum"), Profit=("Profit", "sum"), Orders=("Order ID", "nunique"), Quantity=("Quantity", "sum")).reset_index()
    fig = px.bar(region_summary, x="Region", y=["Sales", "Profit"], barmode="group", title="Sales & Profit by Region", color_discrete_sequence=["#667eea", "#764ba2"], text_auto=",.0f")
    show_fig(fig, height=400)
    col1, col2 = st.columns(2)
    for col, title, container in [("State", "Top 10 States by Sales", col1), ("City", "Top 10 Cities by Sales", col2)]:
        with container:
            top = filtered.groupby(col)["Sales"].sum().sort_values(ascending=False).head(10).reset_index()
            fig = px.bar(top, x="Sales", y=col, orientation="h", title=title, color="Sales", color_continuous_scale="Purples")
            show_fig(fig, height=450, y_reversed=True, showlegend=False)
    st.markdown("---")
    st.subheader("Regional Details")
    st.dataframe(
        region_summary.style.format({"Sales": "${:,.0f}", "Profit": "${:,.0f}", "Orders": "{:,}", "Quantity": "{:,}"}),
        use_container_width=True,
    )

render_filtered_table(filtered)
```

Code Explanation

1 Imports and Page Configuration

The file begins by importing five libraries. pandas handles all data loading and transformation. plotly.express and plotly.graph_objects together cover both quick chart creation (Express) and fine-grained multi-trace figures (Graph Objects). streamlit provides the entire web application framework including widgets, layout containers, and state management. pathlib.Path provides a clean, platform-independent way to check whether the feedback CSV file exists on disk before attempting to read it.

st.set_page_config() must be the very first Streamlit call in any script. It sets the browser tab title to 'Business Insight Dashboard', adds a chart emoji as the favicon, and selects the wide layout so that charts span the full browser width instead of being constrained to a narrow centre column. The session_state initialization checks whether the theme key already exists before setting it, preventing the theme from resetting to light on every user interaction that triggers a script re-run.

2 Hero Banner

The colourful gradient banner at the top of the page is rendered by injecting a raw HTML div using st.markdown with unsafe_allow_html=True. The div uses an inline CSS linear-gradient from violet (#7c3aed) to pink (#ec4899), rounded corners, and a large drop shadow. This purely cosmetic element is used to give the dashboard a modern, branded appearance and to provide the user with an immediate visual anchor when the page loads.

3 Theme Dictionary and BASE_CSS Template

All colour tokens for both the light and dark themes are stored in a Python dictionary called THEMES. Each theme entry holds nine named colour values covering the primary accent colour, secondary accent, main background, secondary background, primary text colour, secondary text colour, border colour, card shadow, and expander background. This centralised dictionary makes it trivial to adjust any colour across the entire application by changing a single value.

BASE_CSS is a multi-line string containing a complete CSS stylesheet written with Python's str.format() placeholder syntax using double braces for literal braces and single braces for variable substitution. When the theme is applied, BASE_CSS.format(**theme) substitutes all colour tokens and the resulting CSS is injected into the page. The stylesheet targets Streamlit's internal data-testid attributes to override the default styling of metric cards, sidebar, buttons, expanders, and data tables. The @import rule at the top loads the Poppins font family from Google Fonts so that the entire UI uses the same modern sans-serif typeface.

4 show_fig() Helper Function

The show_fig() function is a small wrapper around st.plotly_chart(). It accepts a Plotly figure and an arbitrary number of keyword arguments which are forwarded to fig.update_layout(). The template='plotly_white' setting is always applied first, giving every chart a clean white background with light gridlines. The optional y_reversed parameter flips the y-axis direction, which is needed for horizontal bar charts so that the highest-value item appears at the top rather than the bottom. Using this helper eliminates the repetition of three identical lines of boilerplate code after every chart creation call.

5 Column Configuration Dictionaries

Three dictionaries — TEXT_COLS, DATE_COLS, and NUM_COLS — define display metadata for the data table at the bottom of each view. TEXT_COLS maps column names to width hints (small, medium, large) that control how wide each text column appears. DATE_COLS specifies the display format for date columns. NUM_COLS provides format strings for numeric columns: dollar amounts with two decimal places, percentage for discount, and plain integer with comma separator for quantity. These dictionaries are consumed by the render_filtered_table() function.

6 render_filtered_table() Function

This function renders the interactive data table and download button at the bottom of every view. It first creates a display copy of the filtered DataFrame to avoid mutating the analysis data. Date columns are converted to Python date objects so they display as YYYY-MM-DD rather than as full datetime strings with time components. The Postal Code column is cast to string and the

trailing .0 artifact that Pandas adds to integer columns stored as floats is stripped using a regex replacement.

The function uses the USE_COLUMN_CONFIG flag to branch between the modern st.column_config API (available in Streamlit 1.22 and later) and the older style.format() approach for older Streamlit versions. When the column config API is available, it creates typed column objects for text, date, and number columns, which gives the table formatted values, appropriate sort behaviour, and responsive widths. The table height is computed dynamically to show the data rows without excessive empty space or excessive scrolling. A download button is rendered immediately below the table to allow users to export the currently visible filtered data as a CSV file.

7 load_data() and Caching

The @st.cache_data decorator tells Streamlit to store the return value of load_data() in an in-memory cache keyed by the function arguments. On the first call, the CSV is read from disk, dates are parsed, and Year and Month columns are derived. On every subsequent call with the same argument, Streamlit returns the cached DataFrame instantly without re-reading the file. This is critical for performance because Streamlit re-executes the entire script from top to bottom on every user interaction; without caching, the 9,994-row CSV would be reloaded and reprocessed dozens of times per session.

The function accepts an optional uploaded_file argument. When None, it reads the built-in Superstore sample CSV from the local file system. When a file object is passed (from the sidebar file uploader), it reads that file instead, with error handling that returns an empty DataFrame if the file cannot be parsed. After loading, both Order Date and Ship Date are converted to Pandas datetime objects using pd.to_datetime with errors='coerce', which replaces any unparseable values with NaT rather than raising an exception. The Year column is extracted as an integer and the Month column is formatted as a YYYY-MM period string for use in temporal grouping operations.

8 Generic Mode Detection

After loading the data, the code checks whether the three columns required for the specialised analytical views — Sales, Profit, and Order ID — are all present. The any() expression evaluates to True if at least one required column is missing, setting generic_mode to True. In generic mode, the sidebar shows a simplified single-column filter instead of the full four-dimensional filter set, and the main area renders a generic dataset explorer rather than the business-specific views. This design ensures the application degrades gracefully for non-retail datasets rather than crashing with a KeyError.

9 Sidebar Controls

The entire sidebar is constructed within a with st.sidebar block. The Quick Guide expander provides onboarding text for new users. The theme toggle consists of two buttons arranged in a two-column layout; clicking either button updates session_state.theme and calls st.rerun() to force an immediate full-page refresh with the new CSS applied. The four multi-select filters are constructed programmatically using the filter_defs list, which stores each filter's column name, label, warning key, and optional default value function. The options_for() helper safely retrieves unique sorted values from a column, falling back to an empty list with a sidebar warning if the column does not exist. The view selector radio widget allows the user to switch between five named views.

10 Feedback Form and Admin Panel

The feedback form is placed inside a collapsible expander at the bottom of the sidebar. It uses st.form() to batch widget interactions so that the form is only submitted when the user explicitly clicks the Submit button rather than on every slider movement. On submission, a one-row DataFrame containing the timestamp, numeric rating, and optional comment text is appended to feedback.csv using Pandas to_csv in append mode. The header parameter checks whether the file already exists to avoid writing the column header on every submission.

The admin panel is hidden behind a toggle and a PIN check. The admin PIN is read from Streamlit's secrets management system, which keeps sensitive values out of the source code. When a correct PIN is entered, the application reads feedback.csv, sorts it by timestamp in descending order, and

displays the fifty most recent feedback entries in a compact data table. This gives the project team visibility into user feedback without requiring access to the server file system.

11 Filter Application

After the sidebar controls, the filtered DataFrame is constructed by starting from a full copy of df and progressively applying each active filter. For generic mode, only the single user-selected column filter is applied. For the standard mode, four separate boolean index operations are applied in sequence, one for each of Year, Region, Category, and Segment. Each filter is only applied if the user has selected at least one value (non-empty list) and if the corresponding column exists in the DataFrame. This guards against datasets that lack certain columns and provides the 'blank equals all' filter behaviour expected by users.

12 Overview View

The Overview view computes five aggregate metrics from the filtered DataFrame: total sales, total profit, total unique orders, average discount percentage, and overall profit margin. These are displayed in five equally wide metric cards across the top of the page using Streamlit's st.metric() component, which renders each value in a large font inside a gradient-coloured card courtesy of the BASE_CSS overrides. Below the metrics, four charts are arranged in a two-by-two grid: a grouped bar chart comparing Sales and Profit by Category, a donut pie chart showing Sales distribution by Customer Segment, another grouped bar chart for regional Sales and Profit, and a donut pie showing Sales by Ship Mode.

13 Top Products View

The Top Products view allows the user to control the number of displayed products using a slider (range 5 to 20, default 10). Two horizontal bar charts side by side show the top N products ranked by total Sales and by total Profit respectively. Using nlargest(n) on the grouped series efficiently extracts only the required rows without sorting the entire DataFrame. The y_reversed=True argument to show_fig() ensures the highest-value product appears at the top. A sub-category

grouped bar chart below the product charts provides a category-level performance overview with x-axis labels rotated 45 degrees to prevent overlap.

14 Sales Trends View

The Sales Trends view resamples the filtered DataFrame at monthly frequency using Pandas' time-series resampling capability. The monthly aggregation sums both Sales and Profit for each calendar month, producing a two-column DataFrame indexed by month. A Plotly Graph Objects Figure with two Scatter traces (one for Sales in violet, one for Profit in purple) is used rather than Plotly Express because Express does not natively support multi-column line charts from a wide-format DataFrame with a single function call. The yearly and quarterly breakdowns below use standard Plotly Express bar charts with the text_auto parameter to display formatted values directly on the bars.

15 Profit Analysis View

The Profit Analysis view contains four analytical components. Two scatter plots side by side explore Sales vs Profit and Discount vs Profit respectively, with points coloured by product Category. The hover_data parameter on the Sales vs Profit chart adds the Product Name to the tooltip, enabling the user to identify specific products at any point in the chart. The loss-making products horizontal bar chart uses the Reds_r continuous colour scale so that the most negative (loss-making) bars appear in the darkest red. The sub-category profit margin chart uses a Python list comprehension to generate a per-bar colour list, colouring negative-margin bars red and positive-margin bars violet, providing immediate visual identification of loss-making sub-categories.

16 Regional Breakdown View and render_filtered_table

The Regional Breakdown view uses Pandas groupby with named aggregation (the .agg() method with keyword arguments) to compute Sales, Profit, Orders, and Quantity simultaneously for each Region in a single operation. The result is displayed as a grouped bar chart at the top. The top-10 States and Cities charts are constructed by a shared for loop that iterates over column name and container pairs, avoiding code repetition. At the very end of the file, render_filtered_table(filtered)

is called unconditionally so that the interactive data table and download button appear at the bottom of every view.

Dashboard Views Summary

The completed dashboard delivers five fully functional interactive views. The Overview provides an immediate high-level summary of business health through five KPI metrics and four charts covering category, segment, region, and shipping performance. Top Products enables rapid identification of the highest-revenue and highest-profit products as well as the best and worst-performing sub-categories. Sales Trends provides temporal analysis from monthly granularity up through yearly and quarterly views, enabling identification of growth trends and seasonal patterns. Profit Analysis surfaces the profitability dynamics of the business including the discount-profit relationship, loss-making products, and sub-category margin variation. Regional Breakdown presents geographic performance data at region, state, and city levels alongside a summary statistics table. All views share the same filter state, theme, and the interactive data table at the bottom.

Dashboard Views

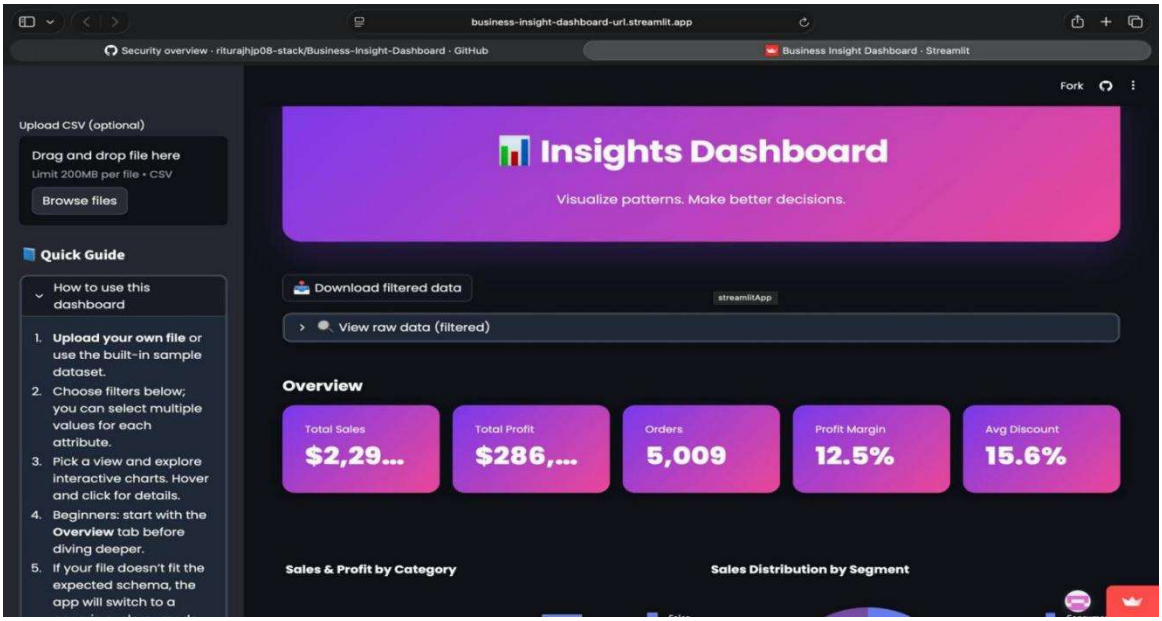

Fig 1:overview

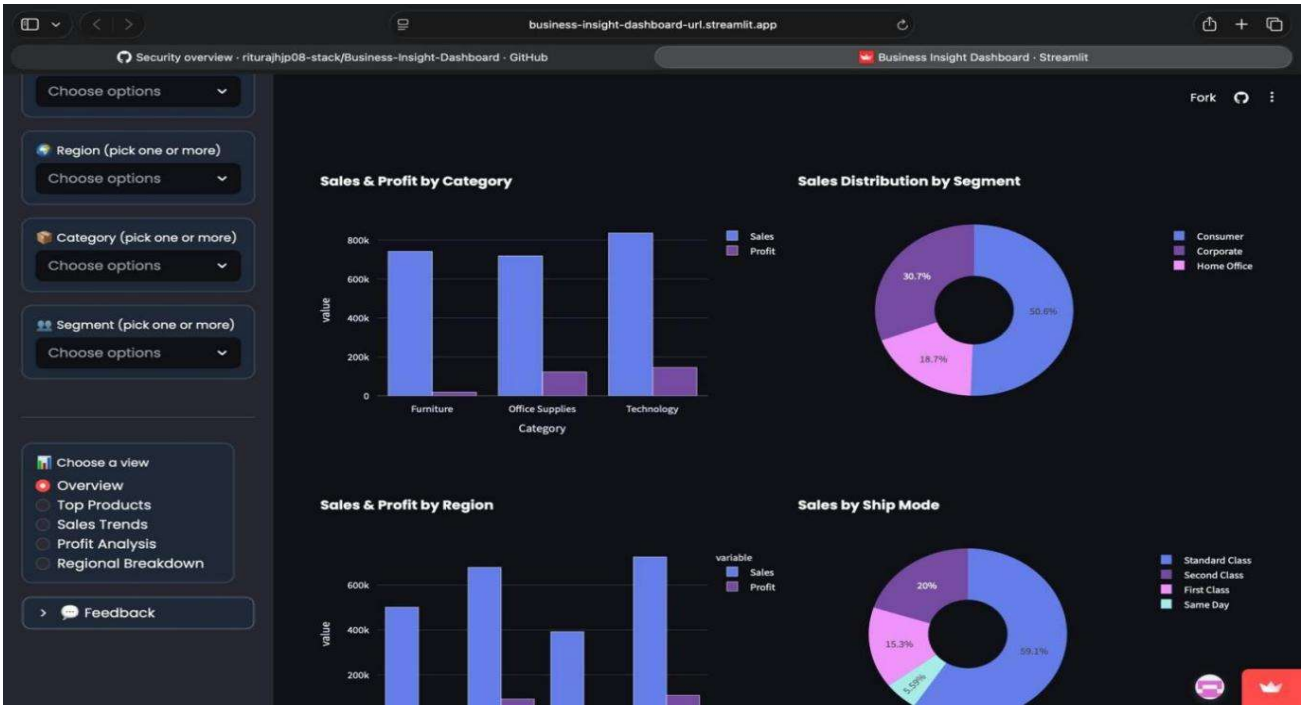

Fig 2:sales and profit

Fig 3:making product

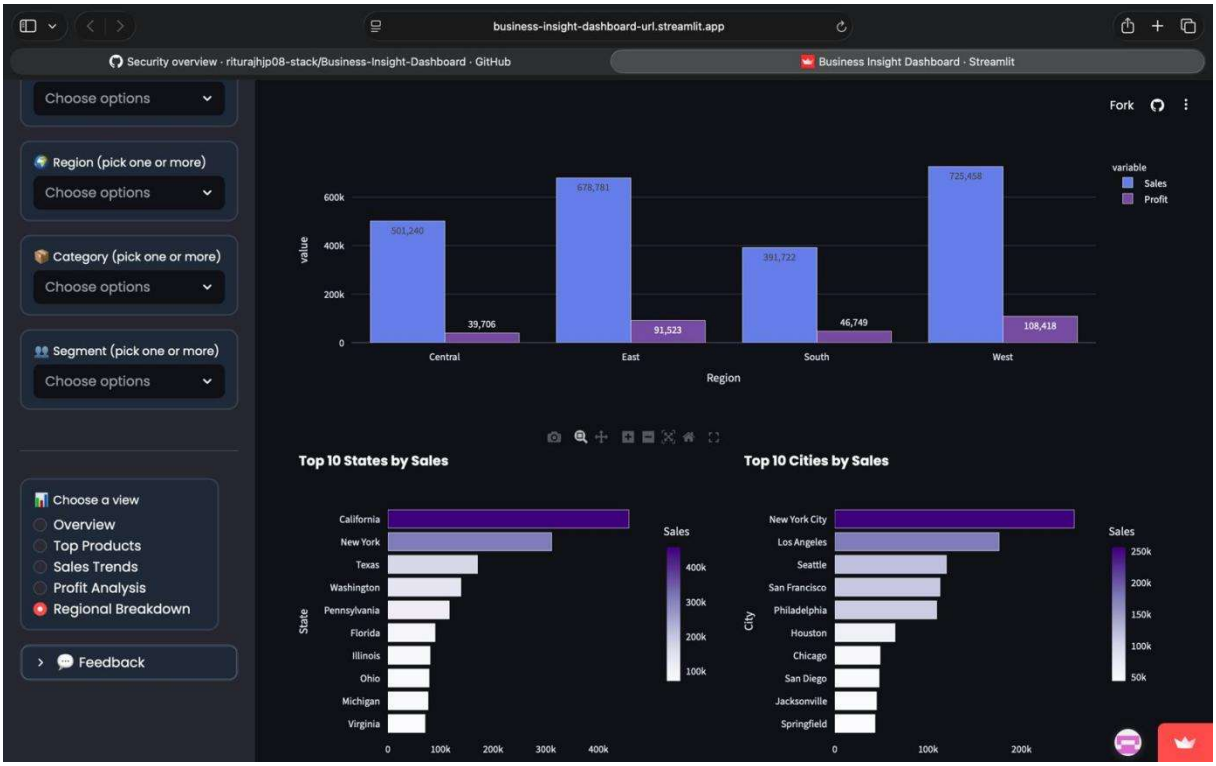

Fig 4:sales category

Goals

Rituraj:

- To develop and implement project modules.
- To handle backend operations efficiently.

Deepanshi:

- To prepare project documentation and report.
- To assist in design and presentation.

Shourya:

- To support basic team coordination.
- To assist when required.

Jiya :

- To contribute to general project support.
- To assist the team when needed.

GitHub link and Streamlit link

GitHub: https://github.com/riturajhjp08-stack/Business-Insight-Dashboard.git

Streamlit : https://business-insight-dashboard-url.streamlit.app/